

# Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit**.

## Vintage mix “ read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

## The one rule behind this layout

**The rung is a property of (CARD, COLUMN) “ never of the column name.** `beta` is edge-rung on the edge card and family-rung on the `gene_family` card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

## How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits “ otherwise `n` and `z` match ordinary prose.

**One correction the measurement needed:** mention-frequency is not importance “ a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

## Cards

- edge “ 186 columns, 45 blocks
- gene “ 135 columns, 29 blocks
- gene\_family “ 61 columns, 12 blocks
- arm “ 296 columns, 43 blocks
- seed\_family “ 33 columns, 2 blocks

# edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer – check the vintage banner before quoting a fitted one.

186 columns 5,649 rows 45 blocks median coverage 72%

## Contents – 45 blocks, most important first

1. (bare) 11 – CORE

2. coupling\_ 13 – CORE

3. seed\_ 1 – CORE

4. arm\_ 15 – CORE

5. adm\_ 4 – CORE

6. beta\_ 8 – CORE

7. n\_ 5 – CORE

8. share\_ 4 – CORE

9. rank\_ 4 – CORE

10. shift\_ 1 – CORE

11. cal\_ 7 – CORE

12. echim\_ 3 – CORE

13. ctx\_ 18 – SUPPORTING

14. healthy\_ 3 – SUPPORTING

15. identity\_ 5 – SUPPORTING

16. dose\_ 7 – SUPPORTING

17. d\_ 4 – SUPPORTING

18. gene\_ 6 – SUPPORTING

19. ago\_ 2 – SUPPORTING

20. pip\_ 2 – SUPPORTING

21. retention\_ 3 – REFERENCE

22. oof\_ 3 – REFERENCE
23. w\_ 1 – REFERENCE

24. own\_ 1 – REFERENCE

25. grank\_ 3 – REFERENCE

26. composition\_ 1 – REFERENCE

27. m\_ 1 – REFERENCE

28. prior\_ 1 – REFERENCE

29. edge\_ 1 – REFERENCE

30. term\_ 3 – REFERENCE

31. mean\_ 2 – REFERENCE

32. surrogate\_ 2 – REFERENCE

33. esub\_ 8 – REFERENCE

34. wiring\_ 1 – REFERENCE

35. kd\_ 2 – REFERENCE

36. realization\_ 2 – REFERENCE

37. cell\_ 1 – REFERENCE

38. sd\_ 1 – REFERENCE

39. acquisition\_ 1 – REFERENCE

40. z\_ 1 – REFERENCE

41. heur\_ 1 – REFERENCE

42. family\_ 1 – REFERENCE

43. cptac\_ 20 – REFERENCE

44. hly\_ 1 – REFERENCE

45. is\_ 1 – REFERENCE

### (BARE) – 11

core – the columns findings rest on

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY\_NAT is the FIELD delta, NAT\_TUM the malignant step.

11 columns – coverage 63% – 100% (median 96%) – 3 share the block description  
4× real, signed – 2× identifier – 2× real – 0 – 2× signed – 1 – 1 – 1× boolean

#### arm

KEY IDENTIFIER 100%

KEY – mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping.

#### beta

EDGE REAL – 0 100%

EDGE-rung coupling coefficient – `readouts.run(level='arm')`, the same-seed collapse REMOVED.

|                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>dGlobal_HLY_NAT</div> <div>ARMREAL, SIGNED63%</div>                                                                                                                                                                                                                                                                  | <div>described by the block above Â· -40  â€" 75</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)</div>         |
| <div>dGlobal_HLY_TUM</div> <div>ARMREAL, SIGNED63%</div>                                                                                                                                                                                                                                                                  | <div>described by the block above Â· -35  â€" 75</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)</div>         |
| <div>dGlobal_NAT_TUM</div> <div>ARMREAL, SIGNED68%</div>                                                                                                                                                                                                                                                                  | <div>described by the block above Â· -49  â€" 74</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                                    |
| <div>dShare_M_own</div> <div>EDGE SIGNED Â·'1Â€ 165%</div>                                                                                                                                                                                                                                                                | <div>Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT.</div>                            |
| <div>dShare_raw_own</div> <div>EDGE SIGNED Â·'1Â€ 165%</div>                                                                                                                                                                                                                                                              | <div>As dShare_M_own but on RAW abundance share, with the model's M weights removed.</div>                                                                                                |
| <div>gene</div> <div>KEYIDENTIFIER100%</div>                                                                                                                                                                                                                                                                              | <div>KEY â€" HGNC gene symbol.</div>                                                                                                                                                      |
| <div>identified</div> <div>EDGEBOOLEAN100%</div>                                                                                                                                                                                                                                                                          | <div> z  &gt; 2 on the UNCALIBRATED posterior SD, where z = beta / beta_sd (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). <b>True on 24.8% of edges.</b></div> |
| <div>identity</div> <div>EDGEREAL, SIGNED96%</div>                                                                                                                                                                                                                                                                        | <div>Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights.</div>                                                                                           |
| <div>z</div> <div>EDGEREAL Â·% 0100%</div>                                                                                                                                                                                                                                                                                | <div>beta / beta_sd for the EDGE-level fit ( readouts.run(level='arm') ), the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD.</div>                          |
| <div>COUPLING_ Â· 13</div> <div>core â€" the columns findings rest on</div>                                                                                                                                                                                                                                               |                                                                                                                                                                                           |
| <div>Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.</div> <div>13 columns Â· coverage 3%â€"100% (median 62%) Â· 9 share the block description</div> <div>5x signed Â·'1Â€ 1 Â· 5x p-value Â· 3x real, signed</div> |                                                                                                                                                                                           |
| <div>coupling_fam</div> <div>FAMILY SIGNED Â·'1Â€ 1100%</div>                                                                                                                                                                                                                                                             | <div>Family-level coupling for the edge's (gene, seed_family) cell.</div>                                                                                                                 |
| <div>coupling_hly</div> <div>EDGE SIGNED Â·'1Â€ 156%</div>                                                                                                                                                                                                                                                                | <div>described by the block above Â· -0.392  â€" +0.36</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)</div>   |
| <div>coupling_hly_surrogate</div> <div>EDGE SIGNED Â·'1Â€ 13%</div>                                                                                                                                                                                                                                                       | <div>Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal.</div>                                                                                  |
| <div>coupling_nat</div> <div>EDGE SIGNED Â·'1Â€ 162%</div>                                                                                                                                                                                                                                                                | <div>described by the block above Â· -0.59  â€" +0.697</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                              |
| <div>coupling_p_hly</div> <div>EDGE P-VALUE56%</div>                                                                                                                                                                                                                                                                      | <div>described by the block above Â· 0  â€" 1</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)</div>            |

|                                                                |                                                                                                                                                                |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>coupling_p_hly_resolved</div> <div>EDGE P-VALUE 59%</div> | <div>described by the block above</div> <div>Defined on: arms with a same-seed SURROGATE (MH-166)</div>                                                        |
| <div>coupling_p_hly_surrogate</div> <div>EDGE P-VALUE 3%</div> | <div>described by the block above</div> <div>Defined on: arms with a same-seed SURROGATE (MH-166)</div>                                                        |
| <div>coupling_p_nat</div> <div>EDGE P-VALUE 62%</div>          | <div>described by the block above</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                        |
| <div>coupling_p_tum</div> <div>EDGE P-VALUE 69%</div>          | <div>p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; &lt;0.05 on 16.2%.</div>                                                |
| <div>coupling_tum</div> <div>EDGE SIGNED 72%</div>             | <div>described by the block above</div> <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div>                         |
| <div>coupling_z_hly</div> <div>EDGE REAL, SIGNED 59%</div>     | <div>described by the block above</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div> |
| <div>coupling_z_nat</div> <div>EDGE REAL, SIGNED 62%</div>     | <div>described by the block above</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                        |
| <div>coupling_z_tum</div> <div>EDGE REAL, SIGNED 69%</div>     | <div>The tumour coupling expressed in NULL SDs the CALIBRATED effect size, and the scale to quote. Median 0.32;  z &gt;2 on 15.5%,  z &gt;3 on 6.4%.</div>     |

SEED\_ 1

core the columns findings rest on

|                                                                           |
|---------------------------------------------------------------------------|
| <div>1 columns coverage 100% (median 100%)</div> <div>1x identifier</div> |
|---------------------------------------------------------------------------|

|                                                          |                                                            |
|----------------------------------------------------------|------------------------------------------------------------|
| <div>seed_family</div> <div>FAMILY IDENTIFIER 100%</div> | <div>KEY the seed family itself, one row per family.</div> |
|----------------------------------------------------------|------------------------------------------------------------|

ARM\_ 15

core the columns findings rest on

|                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|
| <div>15 columns coverage 20% (median 69%)</div> <div>4x real 4x real, signed 3x fraction 1 2x categorical 1x percent 1x boolean</div> |
|---------------------------------------------------------------------------------------------------------------------------------------|

|                                                                    |                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>arm_credit_share</div> <div>ARM-IN-FAMILY FRACTION 20%</div>  | <div>This arm's share of its (gene, seed_family) cell's credit. Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467).</div> <div>Defined on: multi-arm cells only (n_arm_in_cell &gt; 1)</div> |
| <div>arm_dbeta</div> <div>FAMILY REAL 20%</div>                    | <div>max^min across the cell's member arms.</div>                                                                                                                                                                                                                          |
| <div>arm_detection</div> <div>ARM FRACTION 95%</div>               | <div>Fraction of patients in which this ARM is measured at all ( readouts._detection ).</div>                                                                                                                                                                              |
| <div>arm_id_status</div> <div>ARM-IN-FAMILY CATEGORICAL 100%</div> | <div>How the arm's identity within its cell was resolved: resolved_by_dose 2,344 singleton 2,260 unvalidated_balanced 1,045.</div>                                                                                                                                         |

|                                                                          |                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>arm_iqr</div> <div>ARMREAL 69%</div>                                | Interquartile range of the arm's abundance – its DYNAMIC RANGE.                                                                                                                                                                                                    |
| <div>arm_lfc_HLY_NAT_raw</div> <div>ARMREAL, SIGNED63%</div>             | log-FC GTEx-healthy – TCGA-NAT, raw.                                                                                                                                                                                                                               |
| <div>arm_lfc_HLY_TUM_QN</div> <div>ARMREAL, SIGNED64%</div>              | log-FC GTEx-healthy – TCGA-tumour, on quantile-normalised values.                                                                                                                                                                                                  |
| <div>arm_lfc_HLY_TUM_raw</div> <div>ARMREAL, SIGNED63%</div>             | As arm_lfc_HLY_TUM_QN but against the RAW GTEx median.                                                                                                                                                                                                             |
| <div>arm_lfc_NAT_TUM</div> <div>ARMREAL, SIGNED68%</div>                 | log-FC of the arm's median abundance, matched NAT – tumour.                                                                                                                                                                                                        |
| <div>arm_med_rpm</div> <div>ARMREAL 69%</div>                            | Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 – the gap is arms with no expression record, not zeros. <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div> |
| <div>arm_pct_above_floor</div> <div>ARMPERCENT69%</div>                  | Percent of samples in which the arm sits above the detection floor (median 99). Feeds arm_spiker together with arm_iqr. <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div>                                             |
| <div>arm_role_in_family</div> <div>ARM-IN-FAMILYCATEGORICAL72%</div>     | – The ARM's DOSE role inside its seed family – NOT the family's role in the gene, which is what the name suggests.                                                                                                                                                 |
| <div>arm_sep_z</div> <div>FAMILYREAL 20%</div>                           | arm_dbeta in units of the cell's own pooled posterior SD – can these same-seed arms be told apart at all.                                                                                                                                                          |
| <div>arm_share_of_family_dose</div> <div>ARM-IN-FAMILYFRACTION 69%</div> | The arm's share of its seed family's TOTAL dose – denominator is the <b>full family</b> , not the design (median 0.69). The quantity family_role is cut from. <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div>       |
| <div>arm_spiker</div> <div>ARMBOOLEAN69%</div>                           | arm_pct_floor < 40 AND arm_iqr > 1.5 – the arm sits at/below the detection floor in most samples yet swings widely when present.                                                                                                                                   |

ADM\_ 4

core – the columns findings rest on

|                                                                     |                                                                                                                                                                 |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>4 columns – coverage 87%–87% (median 87%)<br/>4× boolean</div> |                                                                                                                                                                 |
| <div>adm_admissible</div> <div>EDGEBOOLEAN87%</div>                 | The admissibility verdict: site + evidence + expression.                                                                                                        |
| <div>adm_expr_gtex</div> <div>EDGEBOOLEAN87%</div>                  | ADMISSIBILITY – could this edge repress in breast AT ALL? has_site (seed match in the 3'UTR) AND expressed (arm present in TCGA tumour OR GTEx healthy breast). |
| <div>adm_expr_tcga</div> <div>EDGEBOOLEAN87%</div>                  | Is the arm expressed above floor in TCGA?                                                                                                                       |
| <div>adm_has_site</div> <div>EDGEBOOLEAN87%</div>                   | Does the arm have a seed match in this target's 3'UTR.                                                                                                          |

BETA\_ 8

core the columns findings rest on

8 columns

coverage 20%100% (median 100%)

4x real 0 4x fraction 01

|                                                            |                                                                                                                                                                                                             |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>beta_arm</div> <div>EDGEREAL 020%</div>               | The arm's OWN $\hat{I}^2$ from the within-cell fit ( <code>arm_rung.py</code> ), defined only in multi-arm cells (fill 0.203).                                                                              |
| <div>beta_deconv</div> <div>EDGEREAL 0100%</div>           | $\hat{I}^2$ refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out).                                                                                        |
| <div>beta_frac</div> <div>EDGEFRACTION 01100%</div>        | MAGNITUDE share ( <code>beta_f</code> / sum beta).                                                                                                                                                          |
| <div>beta_frac_abs</div> <div>EDGEFRACTION 01100%</div>    | $ E[\hat{I}^2_f]  / \hat{I}E[\hat{I}^2]$ the share computed from the POINT ESTIMATES.                                                                                                                       |
| <div>beta_frac_deconv</div> <div>EDGEFRACTION 01100%</div> | <code>beta_frac</code> recomputed on the deconv design the composition-adjusted share of the gene's budget.                                                                                                 |
| <div>beta_frac_sd</div> <div>EDGEFRACTION 01100%</div>     | Posterior SD of the per-draw fraction $\hat{I}^2_f / \hat{I}E\hat{I}^2$ .                                                                                                                                   |
| <div>beta_sd</div> <div>EDGEREAL 0100%</div>               | Posterior summaries of the coupling coefficient from the dense Gibbs fit ( <code>learned/readouts.py</code> ). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta. |
| <div>beta_sd_deconv</div> <div>EDGEREAL 0100%</div>        | Posterior SD on the deconv design.                                                                                                                                                                          |

N\_ 5

core the columns findings rest on

5 columns

coverage 20%100% (median 100%)

4x count 1x constant

|                                                          |                                                                                                                                                          |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>n_HLY_meas</div> <div>EDGECOUNT72%</div>            | How many of this gene's arms have a MEASURED healthy (GTEx) level as opposed to one manufactured by the multi-mapping collapse.                          |
| <div>n_arm_in_cell</div> <div>FAMILYCOUNT20%</div>       | How many arms sit in this edge's (gene, seed_family) CELL i.e. how many of that family's members appear in THIS gene's design.                           |
| <div>n_fam</div> <div>GENECOUNT100%</div>                | How many families compete at this edge's GENE. Gene-rung it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. |
| <div>n_family_in_design</div> <div>FAMILYCOUNT100%</div> | Members of this arm's seed family <b>THAT ARE IN THIS GENE'S DESIGN</b> 1 on <b>4,497 of 5,644 edges (79.7%)</b> .                                       |
| <div>n_samples</div> <div>GENECONSTANT100%</div>         | Samples the edge was fit on CONSTANT (1,040) across the card.                                                                                            |

SHARE\_ 4

core the columns findings rest on

The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms.

4 columns coverage 72% (median 72%) 2 share the block description

4x fraction 1

|                     |                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| share_HLY           | The arm's share of the gene's healthy (GTEx) pressure budget, median 0.129.                                                                                 |
| EDGE FRACTION 1 72% |                                                                                                                                                             |
| share_HLY_meas      | share_HLY over ONLY the arms GTEx genuinely measures invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median 0.244 against the imputed 0.129. |
| EDGE FRACTION 1 46% |                                                                                                                                                             |
| share_NAT           | described by the block above 1                                                                                                                              |
| EDGE FRACTION 1 72% | Defined on: edges with a matched NAT leg (n~104 paired participants)                                                                                        |
| share_TUM           | described by the block above 1                                                                                                                              |
| EDGE FRACTION 1 72% | Defined on: edges present in the progression panel (arm expressed + state measured)                                                                         |

RANK\_ 4

core the columns findings rest on

The arm's WITHIN-GENE rank by dose in one state.

4 columns coverage 72% (median 72%) 4 share the block description

4x count

|                |                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|
| rank_HLY       | described by the block above 1 56                                                                           |
| EDGE COUNT 72% | Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166) |
| rank_HLY_meas  | described by the block above 1 33                                                                           |
| EDGE COUNT 46% | Defined on: edges whose gene has 1 GTEx-MEASURED regulator NaN = GTEx cannot see it (abstention), NOT 0     |
| rank_NAT       | described by the block above 1 58                                                                           |
| EDGE COUNT 72% | Defined on: edges with a matched NAT leg (n~104 paired participants)                                        |
| rank_TUM       | described by the block above 1 59                                                                           |
| EDGE COUNT 72% | Defined on: edges present in the progression panel (arm expressed + state measured)                         |

SHIFT\_ 1

core the columns findings rest on

1 columns coverage 72% (median 72%)

1x categorical

|                      |                                 |
|----------------------|---------------------------------|
| shift_class          | Cross-state class for the edge. |
| EDGE CATEGORICAL 72% |                                 |

CAL\_ 7

core the columns findings rest on

CALIBRATED posterior width ( `learned/calibration.py` ): the reported SD corrected for the measured 1.181.34x narrowness, with `_lo / _hi` spanning the correction constant's own +/-1 SD.

7 columns coverage 100%100% (median 100%) 6 share the block description

4x real 0 3x boolean

|                   |                                                                                                                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cal_beta_sd       | described by the block above 0.00101 3.94                                                                                                                                                                             |
| EDGE REAL 0 100%  |                                                                                                                                                                                                                       |
| cal_beta_sd_hi    | described by the block above 0.00112 4.39                                                                                                                                                                             |
| EDGE REAL 0 100%  |                                                                                                                                                                                                                       |
| cal_beta_sd_lo    | described by the block above 0.000912 3.57                                                                                                                                                                            |
| EDGE REAL 0 100%  |                                                                                                                                                                                                                       |
| cal_identified    | Is the edge identified ( <code> cal_z  &gt; 2</code> ) under the CALIBRATED posterior width the honest version of <code>identified</code> : <b>1,110 vs 1,375 edges</b> , i.e. <b>19.9% [18.122.4]</b> against 24.8%. |
| EDGE BOOLEAN 100% |                                                                                                                                                                                                                       |
| cal_identified_hi | described by the block above 990 true / 4,654 false                                                                                                                                                                   |
| EDGE BOOLEAN 100% |                                                                                                                                                                                                                       |
| cal_identified_lo | described by the block above 1,234 true / 4,410 false                                                                                                                                                                 |
| EDGE BOOLEAN 100% |                                                                                                                                                                                                                       |
| cal_z             | described by the block above 0.386 18.2                                                                                                                                                                               |
| EDGE REAL 0 100%  |                                                                                                                                                                                                                       |

ECHIM\_ 3

core the columns findings rest on

CHIMERIC ligation evidence at its NATIVE (gene, arm) rung a physical duplex read, per source, never pooled.

3 columns coverage 8%26% (median 22%) 2 share the block description

2x count 1x real 0

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| echim_manakov_w | described by the block above 0.125 1.71e+03                                                                   |
| EDGE REAL 0 22% | Defined on: edges with a chimeric (AGO-ligation) duplex aš per-source, NEVER pooled; cross-tissue, not breast |
| echim_n_sources | How many chimeric-eCLIP sources support this edge (1 on 1,151 2 on 262 3 on 34).                              |
| EDGE COUNT 26%  |                                                                                                               |
| echim_tarbase_w | described by the block above 1 28                                                                             |
| EDGE COUNT 8%   | Defined on: edges with a chimeric (AGO-ligation) duplex aš per-source, NEVER pooled; cross-tissue, not breast |

CTX\_ 18

supporting read these when the core raises a question

Gene-design context from the decoy/atlas layer ( learned/analyses/gene\_atlas.py via card\_context ): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

18 columns coverage 84%100% (median 99%) 12 share the block description

4x real 0 4x signed 1 3x real, signed 3x count 2x boolean 1x categorical 1x fraction 01

|                   |                                                                                                                                                                                                                               |      |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| ctx_apriori_class | described by the block above 4 levels                                                                                                                                                                                         |      |
| GENE              | CATEGORICAL                                                                                                                                                                                                                   | 100% |
| ctx_arm_abundant  | Is this ARM above the design's abundance threshold.                                                                                                                                                                           |      |
| ARM               | BOOLEAN                                                                                                                                                                                                                       | 95%  |
| ctx_arm_dose      | The ARM's dose in this gene's design context.                                                                                                                                                                                 |      |
| ARM               | REAL 0                                                                                                                                                                                                                        | 95%  |
| ctx_ceiling       | described by the block above -0.022 +0.44                                                                                                                                                                                     |      |
| GENE              | SIGNED 1                                                                                                                                                                                                                      | 99%  |
| ctx_collapse      | Mean arms collapsed per family in this gene's design (median 1.07) how much the seed-family collapse actually merged. Near 1 means the collapse was almost a no-op for this gene.                                             |      |
| GENE              | REAL 0                                                                                                                                                                                                                        | 100% |
| ctx_d_collin      | Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap.<br>Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns) |      |
| GENE              | SIGNED 1                                                                                                                                                                                                                      | 84%  |
| ctx_d_fam         | Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families).                                                                                                                              |      |
| GENE              | REAL, SIGNED                                                                                                                                                                                                                  | 99%  |
| ctx_dose_max      | described by the block above 0.215 17.8                                                                                                                                                                                       |      |
| GENE              | REAL 0                                                                                                                                                                                                                        | 100% |
| ctx_dose_med      | described by the block above 0.215 17.8                                                                                                                                                                                       |      |
| GENE              | REAL 0                                                                                                                                                                                                                        | 100% |
| ctx_frac_abund    | described by the block above 0 1                                                                                                                                                                                              |      |
| GENE              | FRACTION 1                                                                                                                                                                                                                    | 100% |
| ctx_gap_core      | described by the block above -0.476 +0.541                                                                                                                                                                                    |      |
| GENE              | SIGNED 1                                                                                                                                                                                                                      | 99%  |
| ctx_gap_d_dose    | Dose asymmetry between the real and decoy designs (median 0.43).                                                                                                                                                              |      |
| GENE              | REAL, SIGNED                                                                                                                                                                                                                  | 99%  |
| ctx_gap_deconv    | described by the block above -0.33 +0.374                                                                                                                                                                                     |      |
| GENE              | SIGNED 1                                                                                                                                                                                                                      | 99%  |
| ctx_gap_retention | described by the block above -16.5 17                                                                                                                                                                                         |      |
| GENE              | REAL, SIGNED                                                                                                                                                                                                                  | 95%  |
| ctx_measurable    | described by the block above 4,189 true / 1,428 false                                                                                                                                                                         |      |
| GENE              | BOOLEAN                                                                                                                                                                                                                       | 99%  |
| ctx_n_abund       | described by the block above 0 53                                                                                                                                                                                             |      |
| GENE              | COUNT                                                                                                                                                                                                                         | 100% |

|                                                                                  |                                          |
|----------------------------------------------------------------------------------|------------------------------------------|
| <b>ctx_n_fam_abund</b> <div><div>GENE</div><div>COUNT</div><div>100%</div></div> | <i>described by the block above</i> 0 34 |
| <b>ctx_n_fam_multi</b> <div><div>GENE</div><div>COUNT</div><div>100%</div></div> | <i>described by the block above</i> 0 14 |

**HEALTHY\_ 3**  
supporting read these when the core raises a question

Healthy-state legs and flags including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim.

3 columns coverage 64%72% (median 64%) 3 share the block description  
1x categorical 1x real 1x boolean

|                                                                                        |                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>healthy_leg</b> <div><div>ARM</div><div>CATEGORICAL</div><div>72%</div></div>       | <i>described by the block above</i> 3 levels<br><div>Defined on: arms with a GTE<sub>x</sub>-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>               |
| <b>healthy_potential</b> <div><div>ARM</div><div>REAL 0</div><div>64%</div></div>      | <i>described by the block above</i> 0.27 17.9<br><div>Defined on: arms with a GTE<sub>x</sub>-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>              |
| <b>healthy_uninformative</b> <div><div>ARM</div><div>BOOLEAN</div><div>64%</div></div> | <i>described by the block above</i> 143 true / 3,490 false<br><div>Defined on: arms with a GTE<sub>x</sub>-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div> |

**IDENTITY\_ 5**  
supporting read these when the core raises a question

|                                                                                           |                                                                                                     |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>identity_abs</b> <div><div>EDGE</div><div>FRACTION 01</div><div>96%</div></div>        | identity  as a fraction of the gene's total  identity , <b>clipped to [0,1]</b> .                   |
| <b>identity_allocated</b> <div><div>EDGE</div><div>REAL, SIGNED</div><div>96%</div></div> | <b>Family identity with phantom minor arms forced to 0 the allocation actually used downstream.</b> |
| <b>identity_coherence</b> <div><div>EDGE</div><div>FRACTION 01</div><div>96%</div></div>  | 1 / Î£ identity  over the gene, clipped to (0,1] how much the gene's identity shares CANCEL.        |
| <b>identity_deconv</b> <div><div>EDGE</div><div>REAL, SIGNED</div><div>98%</div></div>    | Shapley identity recomputed under the composition-adjusted design.                                  |
| <b>identity_reliable</b> <div><div>EDGE</div><div>BOOLEAN</div><div>96%</div></div>       | Is this edge's Shapley <code>identity</code> trustworthy identity_coherence 0.5 AND  identity  1?   |

DOSE\_ 7

supporting read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.

7 columns coverage 55%72% (median 72%) 6 share the block description

3x count 2x real 0 1x categorical 1x constant

|                                                            |                                                                                                                                                                     |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>dose_class</div> <div>EDGE CATEGORICAL 72%</div>      | <div>described by the block above 3 levels</div> <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div>                     |
| <div>dose_comp_retention</div> <div>ARM REAL 0 55%</div>   | <div>described by the block above 0.51 1.66</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                   |
| <div>dose_confounded</div> <div>ARM CONSTANT 72%</div>     | <div>Is this edge's NAT's tumour dose a composition/proliferation artifact (either gated retention &lt; 0.4)?</div>                                                 |
| <div>dose_prolif_retention</div> <div>ARM REAL 0 55%</div> | <div>described by the block above 0.69 1.25</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                   |
| <div>dose_rank_HLY</div> <div>EDGE COUNT 72%</div>         | <div>described by the block above 1 53</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div> |
| <div>dose_rank_NAT</div> <div>EDGE COUNT 72%</div>         | <div>described by the block above 1 57</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                        |
| <div>dose_rank_TUM</div> <div>EDGE COUNT 72%</div>         | <div>described by the block above 1 58</div> <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div>                         |

D\_ 4

supporting read these when the core raises a question

Change in the arm's WITHIN-GENE dose rank between two states.

4 columns coverage 46%72% (median 72%) 3 share the block description

4x real, signed

|                                                                 |                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>d_rank_HLY_NAT</div> <div>EDGE REAL, SIGNED 72%</div>      | <div>described by the block above -18 36</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>                                                                                            |
| <div>d_rank_HLY_TUM</div> <div>EDGE REAL, SIGNED 72%</div>      | <div>described by the block above -18 34</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>                                                                                            |
| <div>d_rank_HLY_TUM_meas</div> <div>EDGE REAL, SIGNED 46%</div> | <div>d_rank_HLY_TUM recomputed over arms with a MEASURED healthy level only the collapse-safe version. Prefer it over the bare columnn.</div> <div>Defined on: edges whose gene has 1 GTEx-MEASURED regulator NaN = GTEx cannot see it (abstention), NOT 0</div> |
| <div>d_rank_NAT_TUM</div> <div>EDGE REAL, SIGNED 72%</div>      | <div>described by the block above -12 9</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                                                                                                                    |

GENE\_ 6

supporting read these when the core raises a question

Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.

6 columns coverage 72%100% (median 99%) 5 share the block description  
2x categorical 2x boolean 1x real 0 1x real, signed

|                  |                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------|
| gene_cancer_role | NOT the arm's role the GENE's cancer role ( oncogene / tsg / oncogene/tsg / unknown ), from gene_roles.load_gene_roles . |
| gene_dominated   | described by the block above 2,641 true / 2,854 false                                                                    |
| gene_iqr         | described by the block above 0 6.27<br>Defined on: genes with a NAT-TUM gene-level leg                                   |
| gene_lfc_NAT_TUM | described by the block above -5.85 5.38<br>Defined on: genes with a NAT-TUM gene-level leg                               |
| gene_net_repr    | described by the block above 1,376 true / 4,258 false                                                                    |
| gene_repr_class  | described by the block above 5 levels                                                                                    |

AGO\_ 2

supporting read these when the core raises a question

2 columns coverage 95%100% (median 97%)  
1x fraction 01 1x boolean

|                      |                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ago_loading          | AGO/RISC loading discount.                                                                                                                                                                                  |
| ago_loading_measured | Was ago_loading actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%).<br>Defined on: arms with a Manakov AGO-dominance measurement NOT loading(), which fabricates 1.0 |

PIP\_ 2

supporting read these when the core raises a question

2 columns coverage 100%100% (median 100%)  
1x constant 1x fraction 01

|               |                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------|
| pip_dense     | CONSTANT by construction the coupling readout is called with 1, so every edge has the same value.                            |
| pip_discovery | Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout. |

RETENTION\_ 3

reference context, provenance and rarely-quoted detail

3 columns coverage 72%100% (median 100%)  
1x real 0 1x boolean 1x real, signed

retention\_beta Fraction of an effect surviving covariate adjustment (adjusted / raw).

EDGE REAL 0 100%

retention\_reliable Is this edge's retention ratio trustworthy i.e. does it have a real denominator?

EDGE BOOLEAN 100%

retention\_rho I\_deconv / I\_core

EDGE REAL, SIGNED 72%

OOF\_ 3

reference context, provenance and rarely-quoted detail

Out-of-fold predictive correlation, arm-level vs family-level, and their difference (oof\_drho = arm minus family). Negative = the arm-resolved fit predicts repression better.  
3 columns coverage 20%20% (median 20%) 2 share the block description  
3x signed -11

oof\_drho Arm-level OOF rho MINUS family-level.

FAMILY SIGNED -11  
20%

oof\_rho\_arm described by the block above -0.542 +0.324

FAMILY SIGNED -11  
20% | Defined on: multi-arm cells only (n\_arm\_in\_cell > 1) 20.3% of edges

oof\_rho\_fam described by the block above -0.545 +0.309

FAMILY SIGNED -11  
20% | Defined on: multi-arm cells only (n\_arm\_in\_cell > 1) 20.3% of edges

W\_ 1

reference context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)  
1x real 0

w\_max Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat.

GENE REAL 0 100%

OWN\_ 1

reference context, provenance and rarely-quoted detail

1 columns coverage 88%88% (median 88%)  
1x real 0

own\_specific\_frac

Fraction of the arm's variance that is target-specific.

ARM

REAL  $\hat{\gamma}_0$

88%

GRANK\_  $\hat{\gamma}_3$

reference  $\hat{\gamma}$  context, provenance and rarely-quoted detail

The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.

3 columns

$\hat{\gamma}$  coverage ~~63%~~69% (median 68%)

$\hat{\gamma}_3$  share the block description

3x count

grank\_HLY

ARM

COUNT

63%

described by the block above  $\hat{\gamma}_24$   $\hat{\gamma}$  100

Defined on: arms with a GTE~~x~~-HEALTHY leg (the thinnest state  $\hat{\gamma}$  many arms are multi-mapping-collapsed, MH-166)

grank\_NAT

ARM

COUNT

68%

described by the block above  $\hat{\gamma}_0$   $\hat{\gamma}$  100

Defined on: edges with a matched NAT leg (n~104 paired participants)

grank\_TUM

ARM

COUNT

69%

described by the block above  $\hat{\gamma}_2$   $\hat{\gamma}$  100

Defined on: edges present in the progression panel (arm expressed + state measured)

COMPOSITION\_  $\hat{\gamma}_1$

reference  $\hat{\gamma}$  context, provenance and rarely-quoted detail

1 columns

$\hat{\gamma}$  coverage ~~100%~~100% (median 100%)

1x categorical

composition\_class

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block).

EDGE

CATEGORICAL

100%

M\_  $\hat{\gamma}_1$

reference  $\hat{\gamma}$  context, provenance and rarely-quoted detail

1 columns

$\hat{\gamma}$  coverage ~~100%~~100% (median 100%)

1x real  $\hat{\gamma}_0$

m\_nnls

Bagged-NNLS weight. Exactly 0 in 31.7% of families  $\hat{\gamma}$  this is what lets `identity` say a regulator contributes nothing, which `beta` structurally cannot.

EDGE

REAL  $\hat{\gamma}_0$

100%

PRIOR\_  $\hat{\gamma}_1$

reference  $\hat{\gamma}$  context, provenance and rarely-quoted detail

1 columns

$\hat{\gamma}$  coverage ~~100%~~100% (median 100%)

1x constant

prior\_pi

The Gibbs sampler's inclusion prior  $\hat{\gamma}$  **CONSTANT 0.05** across the card.

EDGE

CONSTANT

100%

EDGE\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 73% (median 73%)  
1x signed 1

edge\_rho\_adj

An edge-rung quantity carried onto this row.

EDGE

SIGNED 1

73%

Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

TERM\_ 3

reference " context, provenance and rarely-quoted detail

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms.

3 columns coverage 68% (median 68%) 3 share the block description  
2x signed 1 1x real, signed

term\_ABUND

described by the block above -0.641 +0.349

EDGE

SIGNED 1

68%

Defined on: edges present in the progression panel (arm expressed + state measured)

term\_INTERACT

described by the block above -0.539 +0.629

EDGE

SIGNED 1

68%

Defined on: edges present in the progression panel (arm expressed + state measured)

term\_WIRING

described by the block above -5.56 3.67

EDGE

REAL, SIGNED

68%

Defined on: edges present in the progression panel (arm expressed + state measured)

MEAN\_ 2

reference " context, provenance and rarely-quoted detail

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises.

2 columns coverage 90% (median 90%) 2 share the block description  
2x real, signed

mean\_dGlobalRank

described by the block above -30.4 42.7

ARM

REAL, SIGNED

90%

Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

mean\_own\_shift

described by the block above -3.94 5.05

ARM

REAL, SIGNED

90%

Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

SURROGATE\_ 2

reference " context, provenance and rarely-quoted detail

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for.

2 columns coverage 3% (median 3%) 2 share the block description  
1x fraction 0 1x categorical

surrogate\_corr

described by the block above 0.541 0.892

ARM

FRACTION 0

3%

Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges

surrogate\_instrument

described by the block above 7 levels

ARM

CATEGORICAL

3%

Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges

ESUB\_ 8

reference “ context, provenance and rarely-quoted detail

8 columns coverage 98%–100% (median 99%) 5 share the block description

6x signed 1 1x categorical 1x p-value

|                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>esub_best_sub</div> <div><div>EDGE</div><div>CATEGORICAL</div><div>100%</div></div> <div>described by the block above 4 levels</div> <div>Defined on: edges with a per-PAM50 heterogeneity fit “ š DISTRIBUTIONAL (MH-165), not a per-edge subtype label</div>       |
| <div>esub_q_adj</div> <div><div>EDGE</div><div>P-VALUE</div><div>98%</div></div> <div>BH-adjusted q for the best-subtype contrast “</div>                                                                                                                                 |
| <div>esub_rho_Basal</div> <div><div>EDGE</div><div>SIGNED 1 1</div><div>99%</div></div> <div>described by the block above -0.555 “ +0.474</div> <div>Defined on: edges with a per-PAM50 heterogeneity fit “ š DISTRIBUTIONAL (MH-165), not a per-edge subtype label</div> |
| <div>esub_rho_Her2</div> <div><div>EDGE</div><div>SIGNED 1 1</div><div>98%</div></div> <div>described by the block above -0.519 “ +0.584</div> <div>Defined on: edges with a per-PAM50 heterogeneity fit “ š DISTRIBUTIONAL (MH-165), not a per-edge subtype label</div>  |
| <div>esub_rho_LumA</div> <div><div>EDGE</div><div>SIGNED 1 1</div><div>99%</div></div> <div>described by the block above -0.387 “ +0.428</div> <div>Defined on: edges with a per-PAM50 heterogeneity fit “ š DISTRIBUTIONAL (MH-165), not a per-edge subtype label</div>  |
| <div>esub_rho_LumB</div> <div><div>EDGE</div><div>SIGNED 1 1</div><div>99%</div></div> <div>described by the block above -0.407 “ +0.429</div> <div>Defined on: edges with a per-PAM50 heterogeneity fit “ š DISTRIBUTIONAL (MH-165), not a per-edge subtype label</div>  |
| <div>esub_rho_best</div> <div><div>EDGE</div><div>SIGNED 1 1</div><div>100%</div></div> <div>Coupling in the arm's BEST PAM50 subtype.</div>                                                                                                                              |
| <div>esub_rho_pool</div> <div><div>EDGE</div><div>SIGNED 1 1</div><div>100%</div></div> <div>Coupling pooled across subtypes “ the unselected reference for esub_rho_best (median 0.0135, indistinguishable from a random-subtype draw at 0.0123).</div>                  |

WIRING\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 68%–68% (median 68%)

1x fraction 0–1

|                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>wiring_frac</div> <div><div>EDGE</div><div>FRACTION 0–1</div><div>68%</div></div> <div>Fraction of the acquisition score attributable to WIRING rather than abundance.</div> <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

KD\_ 2

reference “ context, provenance and rarely-quoted detail

2 columns coverage 85%–85% (median 85%)

1x percent 1x real, signed

|                 |                                                                                                                              |     |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|-----|
| kd_affinity_pct | Is this gene among the arm's strongest predicted targets – a WITHIN-ARM percentile, <b>higher</b> = <b>stronger target</b> . |     |
| EDGE            | PERCENT                                                                                                                      | 85% |
| kd_repression   | scanMiR RBNS binding affinity.                                                                                               |     |
| EDGE            | REAL, SIGNED                                                                                                                 | 85% |

REALIZATION\_ 2

reference context, provenance and rarely-quoted detail

|                                                                                            |
|--------------------------------------------------------------------------------------------|
| Within-patient realization quantities the score, its identity allocation, and who owns it. |
| 2 columns coverage 58%69% (median 63%) 2 share the block description                       |
| 1x fraction 01 1x real, signed                                                             |

|                             |                                                                   |                |                                                                                     |
|-----------------------------|-------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|
| <b>realization_identity</b> | described by the block above $\hat{A} \cdot 0 \hat{A} = 1$        |                |                                                                                     |
| <div>EDGE</div>             | <div>FRACTION 0<math>\hat{A}</math><math>\hat{A}</math>1</div>    | <div>58%</div> | Defined on: genes with a family-level realization fit                               |
| <b>realization_score</b>    | described by the block above $\hat{A} \cdot -4.55 \hat{A} = 6.42$ |                |                                                                                     |
| <div>EDGE</div>             | <div>REAL, SIGNED</div>                                           | <div>69%</div> | Defined on: edges present in the progression panel (arm expressed + state measured) |

CELL\_ 1

reference context, provenance and rarely-quoted detail

|                                        |
|----------------------------------------|
| 1 columns coverage 20%20% (median 20%) |
| 1x boolean                             |

|                      |         |                                                                                                                     |
|----------------------|---------|---------------------------------------------------------------------------------------------------------------------|
| cell_arms_resolvable |         | Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. |
| FAMILY               | BOOLEAN | 20%                                                                                                                 |

SD\_ 1

reference context, provenance and rarely-quoted detail

|                                        |
|----------------------------------------|
| 1 columns coverage 20%20% (median 20%) |
| 1x real 0                              |

|        |                                                |     |
|--------|------------------------------------------------|-----|
| sd_arm | Posterior SD of the arm-level $\hat{\tau}^2$ . |     |
| EDGE   | REAL $\hat{\tau}^2 = 0$                        | 20% |

ACQUISITION\_ 1

reference context, provenance and rarely-quoted detail

|                                        |
|----------------------------------------|
| 1 columns coverage 59%59% (median 59%) |
| 1x real, signed                        |

|                   |                                            |     |                                                                                                                           |
|-------------------|--------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------|
| acquisition_score | Composite acquisition score across states. |     |                                                                                                                           |
| EDGE              | REAL, SIGNED                               | 59% | Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{a} \in$ many arms are multi-mapping-collapsed, MH-166) |

Z\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 20%20% (median 20%)  
1x real 0

z\_arm

beta\_arm / sd\_arm from the WITHIN-CELL fit ( arm\_rung.py ) “ the arm's z inside its (gene, seed\_family) cell only, defined on the 20.3% of edges in multi-arm cells.

EDGE REAL 0 20%

HEUR\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)  
1x count

heur\_gene\_nreg

Regulator count for this gene from the 6b-RETIRED heuristic lane.

GENE COUNT 100%

FAMILY\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 69%69% (median 69%)  
1x signed 11

family\_rho\_adj

Family-level properties carried onto this row (size, the arm's dose share of it, and its role).

FAMILY SIGNED 11 69%

CPTAC\_ 20

reference “ context, provenance and rarely-quoted detail

CPTAC protein channel.

20 columns coverage 45%79% (median 65%) 16 share the block description  
12x signed 11 4x real, signed 2x constant 2x categorical

cptac\_prosp\_comp\_class\_prot

Composition class of this edge's PROTEIN coupling (cell\_intrinsic / partial / composition\_explained) in the prospective cohort.

EDGE CATEGORICAL 59%

cptac\_prosp\_n

Patients in the prospective CPTAC cohort “ CONSTANT 101.

EDGE CONSTANT 79%

cptac\_prosp\_ret\_prot

described by the block above -6.53 “ 6.66  
| Defined on: genes/edges present in the CPTAC cohort

EDGE REAL, SIGNED 59%

cptac\_prosp\_ret\_rna

described by the block above -3.22 “ 5.31  
| Defined on: genes/edges present in the CPTAC cohort

EDGE REAL, SIGNED 61%

cptac\_prosp\_rho\_disc

described by the block above -0.729 “ +0.56  
| Defined on: genes/edges present in the CPTAC cohort

EDGE SIGNED 11 78%

|                                                                                                                          |                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>cptac_prosp_rho_disc_raw</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 78%</div></div> | <div>described by the block above <math>\hat{A} \cdot -0.698 \hat{A} \in +0.682</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_rho_prot</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 78%</div></div>     | <div>described by the block above <math>\hat{A} \cdot -0.721 \hat{A} \in +0.564</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_rho_prot_raw</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 79%</div></div> | UNADJUSTED protein coupling, prospective.                                                                                                                   |
| <div><div>cptac_prosp_rho_rna</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 79%</div></div>      | <div>described by the block above <math>\hat{A} \cdot -0.509 \hat{A} \in +0.611</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_rho_rna_raw</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 79%</div></div>  | <div>described by the block above <math>\hat{A} \cdot -0.578 \hat{A} \in +0.692</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_comp_class_prot</div><div>EDGE CATEGORICAL 45%</div></div>                                          | <div>described by the block above <math>\hat{A} \cdot 3 \text{ levels}</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div>          |
| <div><div>cptac_t105_n</div><div>EDGE CONSTANT 70%</div></div>                                                           | Patients in the TCGA-overlap CPTAC cohort $\hat{A} \in$ <b>CONSTANT 105</b> . See the prospective twin.                                                     |
| <div><div>cptac_t105_ret_prot</div><div>EDGE REAL, SIGNED 45%</div></div>                                                | <div>described by the block above <math>\hat{A} \cdot -5.78 \hat{A} \in 4.86</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div>    |
| <div><div>cptac_t105_ret_rna</div><div>EDGE REAL, SIGNED 49%</div></div>                                                 | <div>described by the block above <math>\hat{A} \cdot -4.93 \hat{A} \in 5.96</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div>    |
| <div><div>cptac_t105_rho_disc</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 64%</div></div>      | <div>described by the block above <math>\hat{A} \cdot -0.587 \hat{A} \in +0.675</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_rho_disc_raw</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 65%</div></div>  | <div>described by the block above <math>\hat{A} \cdot -0.643 \hat{A} \in +0.727</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_rho_prot</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 64%</div></div>      | <div>described by the block above <math>\hat{A} \cdot -0.52 \hat{A} \in +0.671</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div>  |
| <div><div>cptac_t105_rho_prot_raw</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 65%</div></div>  | <div>described by the block above <math>\hat{A} \cdot -0.714 \hat{A} \in +0.736</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_rho_rna</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 64%</div></div>       | <div>described by the block above <math>\hat{A} \cdot -0.593 \hat{A} \in +0.684</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_rho_rna_raw</div><div>EDGE SIGNED <math>\hat{A}^{\prime 1} \hat{A} \in \{1</math> 65%</div></div>   | <div>described by the block above <math>\hat{A} \cdot -0.782 \hat{A} \in +0.695</math></div> <div>Defined on: genes/edges present in the CPTAC cohort</div> |

HLY\_  $\hat{A} \cdot 1$

reference  $\hat{A} \in$  context, provenance and rarely-quoted detail

1 columns  $\hat{A} \cdot$  coverage **61 $\hat{A} \in$ 61%** (median 61%)

1x boolean

|                                                                     |                                                                                                                                                                                        |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>hly_leg_concordant</div><div>EDGE BOOLEAN 61%</div></div> | Do the two HLY $\hat{A} \cdot$ TUM legs ( <code>_QN</code> and <code>_raw</code> ) at least AGREE ON SIGN. <b>Concordant 76.1%, DISCORDANT 23.9%</b> of the 3,433 edges carrying both. |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

IS\_ 1  
reference " context, provenance and rarely-quoted detail

|    |         |          |                     |
|----|---------|----------|---------------------|
| 1  | columns | coverage | 58%58% (median 58%) |
| 1x | boolean |          |                     |

is\_realization\_owner

EDGEBOOLEAN58%

A boolean flag; read the suffix.  
Defined on: genes with a family-level realization fit

realization/gene\_card.tsv

# gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance – `agg_of` carries the meaning instead.

135 columns 1,549 rows 29 blocks median coverage 83%

## Contents – 29 blocks, most important first

1. n\_ 7 – CORE

2. (bare) 2 – CORE

3. comp\_ 16 – CORE

4. ctx\_ 16 – CORE

5. top\_ 8 – SUPPORTING

6. realized\_ 5 – SUPPORTING

7. w\_ 1 – SUPPORTING

8. lit\_ 7 – SUPPORTING

9. cptac\_ 24 – REFERENCE

10. dominant\_ 4 – REFERENCE

11. frac\_ 3 – REFERENCE

12. static\_ 1 – REFERENCE

13. total\_ 1 – REFERENCE

14. owner\_ 1 – REFERENCE
15. median\_ 1 – REFERENCE

16. armres\_ 7 – REFERENCE

17. identity\_ 1 – REFERENCE

18. heur\_ 5 – REFERENCE

19. greal\_ 10 – REFERENCE

20. acquired\_ 2 – REFERENCE

21. pressure\_ 1 – REFERENCE

22. realization\_ 1 – REFERENCE

23. tcga\_ 4 – REFERENCE

24. disc\_ 2 – REFERENCE

25. max\_ 1 – REFERENCE

26. regulatory\_ 1 – REFERENCE

27. mean\_ 1 – REFERENCE

28. concentration\_ 1 – REFERENCE

29. fam\_ 1 – REFERENCE

## N\_ – 7

core – the columns findings rest on

7 columns – coverage 91%–100% (median 100%)  
7× count

### n\_arms

GENE COUNT AGG ARM 100%

Arms in the LEARNED MODEL's design – `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90).

### n\_cell\_intrinsic

GENE COUNT 100%

FAMILIES whose coupling SURVIVES the composition block (retention ≥ 0.7).

### n\_composition\_explained

GENE COUNT 100%

FAMILIES whose coupling does NOT survive the composition block (retention < 0.4).

### n\_fam

GENE COUNT AGG FAMILY  
100%

How many seed families regulate this gene.

### n\_hallmark\_sets

GENE COUNT 91%

How many MSigDB Hallmark programs contain this gene (1–10, median 2). Membership only – it says nothing about regulation.

### n\_identified

GENE COUNT 100%

Families reaching  $|z| > 2$  on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** – meaning the model cannot distinguish ANY of that gene's families from zero.

|                                                                                                                                                                                                                                                            |                                |                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n_pip_disc_gt50                                                                                                                                                                                                                                            |                                | Count of the gene's families with <code>pip_discovery &gt; 0.5</code> under the evidence- $\hat{I}$ readout.                                                                                                                                                                                                       |
| GENE                                                                                                                                                                                                                                                       | COUNT                          | 100%                                                                                                                                                                                                                                                                                                               |
| (BARE) $\hat{A}$ 2                                                                                                                                                                                                                                         |                                |                                                                                                                                                                                                                                                                                                                    |
| core $\hat{A}$ the columns findings rest on                                                                                                                                                                                                                |                                |                                                                                                                                                                                                                                                                                                                    |
| 2 columns $\hat{A}$ coverage 100% $\hat{A}$ 100% (median 100%)<br>1 $\times$ fraction 0 $\hat{A}$ 1 $\hat{A}$ 1 $\times$ identifier                                                                                                                        |                                |                                                                                                                                                                                                                                                                                                                    |
| concentration                                                                                                                                                                                                                                              |                                | <code>beta.max() / beta.sum()</code> $\hat{A}$ the TOP FAMILY's share of the gene's total $\hat{I}^2$ ; median 0.954, i.e. most genes are dominated by one family.                                                                                                                                                 |
| GENE                                                                                                                                                                                                                                                       | FRACTION 0 $\hat{A}$ 1         | 100%                                                                                                                                                                                                                                                                                                               |
| gene                                                                                                                                                                                                                                                       |                                | KEY $\hat{A}$ HGNC gene symbol.                                                                                                                                                                                                                                                                                    |
| KEY                                                                                                                                                                                                                                                        | IDENTIFIER                     | 100%                                                                                                                                                                                                                                                                                                               |
| COMP_ $\hat{A}$ 16                                                                                                                                                                                                                                         |                                |                                                                                                                                                                                                                                                                                                                    |
| core $\hat{A}$ the columns findings rest on                                                                                                                                                                                                                |                                |                                                                                                                                                                                                                                                                                                                    |
| Cell-composition drivers of this gene's coupling ( <code>compartment_*</code> ): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.                                                                       |                                |                                                                                                                                                                                                                                                                                                                    |
| 16 columns $\hat{A}$ coverage 59% $\hat{A}$ 90% (median 73%) $\hat{A}$ 7 share the block description<br>6 $\times$ categorical $\hat{A}$ 6 $\times$ signed $\hat{A}$ 1 $\hat{A}$ 1 $\hat{A}$ 3 $\times$ real, signed $\hat{A}$ 1 $\times$ real $\hat{A}$ 0 |                                |                                                                                                                                                                                                                                                                                                                    |
| comp_cptac_prot_driver                                                                                                                                                                                                                                     |                                | described by the block above $\hat{A}$ 10 levels<br><div>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</div>                                                                                                                                                                  |
| GENE                                                                                                                                                                                                                                                       | CATEGORICAL                    | 59%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_driver_ret                                                                                                                                                                                                                                 |                                | 1 $\hat{A}$ <code>comp_cptac_prot_driver_share</code> , exactly. Median +0.398 raw, +0.469 gated to <code> i_raw  <math>\hat{A}</math>%</code> $\hat{A}$ 0.10 $\hat{A}$ quote the gated value. See the mRNA twin.<br><div>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</div> |
| GENE                                                                                                                                                                                                                                                       | REAL, SIGNED                   | 59%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_driver_share                                                                                                                                                                                                                               |                                | Protein-layer twin of <code>comp_tcga_mrna_driver_share</code> . Same exact complementarity with <code>_driver_ret</code> (8.9e-16 on 918 rows).                                                                                                                                                                   |
| GENE                                                                                                                                                                                                                                                       | REAL, SIGNED                   | 59%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_rho_raw                                                                                                                                                                                                                                    |                                | Unadjusted aggregate PROTEIN coupling $\hat{A}$ denominator of <code>comp_cptac_prot_driver_ret</code> .                                                                                                                                                                                                           |
| GENE                                                                                                                                                                                                                                                       | SIGNED $\hat{A}$ 1 $\hat{A}$ 1 | 73%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_top_budget                                                                                                                                                                                                                                 |                                | described by the block above $\hat{A}$ 10 levels<br><div>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</div>                                                                                                                                                                  |
| GENE                                                                                                                                                                                                                                                       | CATEGORICAL                    | 73%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_top_budget_r                                                                                                                                                                                                                               |                                | described by the block above $\hat{A}$ -0.671 $\hat{A}$ +0.72<br><div>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</div>                                                                                                                                                     |
| GENE                                                                                                                                                                                                                                                       | SIGNED $\hat{A}$ 1 $\hat{A}$ 1 | 73%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_top_target                                                                                                                                                                                                                                 |                                | described by the block above $\hat{A}$ 10 levels<br><div>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</div>                                                                                                                                                                  |
| GENE                                                                                                                                                                                                                                                       | CATEGORICAL                    | 73%                                                                                                                                                                                                                                                                                                                |
| comp_cptac_prot_top_target_r                                                                                                                                                                                                                               |                                | described by the block above $\hat{A}$ -0.745 $\hat{A}$ +0.842<br><div>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</div>                                                                                                                                                    |
| GENE                                                                                                                                                                                                                                                       | SIGNED $\hat{A}$ 1 $\hat{A}$ 1 | 73%                                                                                                                                                                                                                                                                                                                |
| comp_tcga_mrna_driver                                                                                                                                                                                                                                      |                                | WHICH compartment drives the gene's mRNA coupling $\hat{A}$ the one whose removal moves it most (CAF <sub>s</sub> 279 $\hat{A}$ purity 174 $\hat{A}$ prolif 148 $\hat{A}$ Endothelial 122 $\hat{A}$ Myeloid 109).                                                                                                  |
| GENE                                                                                                                                                                                                                                                       | CATEGORICAL                    | 66%                                                                                                                                                                                                                                                                                                                |

|                                                                                                                |                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>comp_tcga_mrna_driver_ret</div> <div>GENE REAL, SIGNED 66%</div>                                          | 1 $\hat{A}^{\prime}$ comp_tcga_mrna_driver_share, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment.                                                                                                                                   |
| <div>comp_tcga_mrna_driver_share</div> <div>GENE REAL <math>\hat{A}^{\prime} \hat{A} \leq 0</math> 66%</div>   | Share of the gene's mRNA coupling attributable to the single compartment that most changes it $\hat{A}^{\prime}$ <b>axiom 8's composition GATE</b> , the axis that separates the harm tails at $p=0.00999$ where the ensemble axis gives $p=0.10$ .                                                         |
| <div>comp_tcga_mrna_rho_raw</div> <div>GENE SIGNED <math>\hat{A}^{\prime} \hat{A} \leq 1</math> 90%</div>      | The gene's UNADJUSTED aggregate mRNA coupling $\hat{A}^{\prime}$ the DENOMINATOR of comp_tcga_mrna_driver_ret.                                                                                                                                                                                              |
| <div>comp_tcga_mrna_top_budget</div> <div>GENE CATEGORICAL 90%</div>                                           | The compartment the gene's miRNA BUDGET loads on most $\hat{A}^{\prime}$ a property of the REGULATORS.                                                                                                                                                                                                      |
| <div>comp_tcga_mrna_top_budget_r</div> <div>GENE SIGNED <math>\hat{A}^{\prime} \hat{A} \leq 1</math> 90%</div> | <div>described by the block above <math>\hat{A}^{\prime} -0.561 \hat{A} \leq +0.53</math></div> <div>Defined on: genes/families with a TCGA-mRNA compartment driver <math>\hat{A}^{\prime}</math> <math>\hat{A}^{\prime}</math> MH-201: these axes predict the SIGN of <math>\hat{I}^2</math>'s gain</div>  |
| <div>comp_tcga_mrna_top_target</div> <div>GENE CATEGORICAL 90%</div>                                           | The compartment the TARGET GENE's expression loads on most.                                                                                                                                                                                                                                                 |
| <div>comp_tcga_mrna_top_target_r</div> <div>GENE SIGNED <math>\hat{A}^{\prime} \hat{A} \leq 1</math> 90%</div> | <div>described by the block above <math>\hat{A}^{\prime} -0.691 \hat{A} \leq +0.887</math></div> <div>Defined on: genes/families with a TCGA-mRNA compartment driver <math>\hat{A}^{\prime}</math> <math>\hat{A}^{\prime}</math> MH-201: these axes predict the SIGN of <math>\hat{I}^2</math>'s gain</div> |

CTX\_  $\hat{A}^{\prime}$  16

core  $\hat{A}^{\prime}$  the columns findings rest on

Gene-design context from the decoy/atlas layer ( learned/analyses/gene\_atlas.py via card\_context ): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

16 columns  $\hat{A}^{\prime}$  coverage 45% $\hat{A} \leq 100\%$  (median 90%)  $\hat{A}^{\prime}$  13 share the block description

4x signed  $\hat{A}^{\prime} \hat{A} \leq 1$   $\hat{A}^{\prime}$  3x real  $\hat{A}^{\prime} \hat{A} \leq 0$   $\hat{A}^{\prime}$  3x real, signed  $\hat{A}^{\prime}$  3x count  $\hat{A}^{\prime}$  1x categorical  $\hat{A}^{\prime}$  1x fraction 0 $\hat{A} \leq 1$   $\hat{A}^{\prime}$  1x boolean

|                                                                                                 |                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>ctx_apriori_class</div> <div>GENE CATEGORICAL 100%</div>                                   | A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED).                                                                                                               |
| <div>ctx_ceiling</div> <div>GENE SIGNED <math>\hat{A}^{\prime} \hat{A} \leq 1</math> 98%</div>  | <div>described by the block above <math>\hat{A}^{\prime} -0.022 \hat{A} \leq +0.44</math></div>                                                                                                               |
| <div>ctx_collapse</div> <div>GENE REAL <math>\hat{A}^{\prime} \hat{A} \leq 0</math> 90%</div>   | <div>described by the block above <math>\hat{A}^{\prime} 1 \hat{A} \leq 5</math></div>                                                                                                                        |
| <div>ctx_d_collin</div> <div>GENE SIGNED <math>\hat{A}^{\prime} \hat{A} \leq 1</math> 45%</div> | <div>described by the block above <math>\hat{A}^{\prime} -0.699 \hat{A} \leq +0.495</math></div> <div>Defined on: genes with <math>\geq 2</math> seed families (a collinearity measure needs 2 columns)</div> |
| <div>ctx_d_fam</div> <div>GENE REAL, SIGNED 87%</div>                                           | <div>described by the block above <math>\hat{A}^{\prime} -2 \hat{A} \leq 25</math></div>                                                                                                                      |
| <div>ctx_dose_max</div> <div>GENE REAL <math>\hat{A}^{\prime} \hat{A} \leq 0</math> 90%</div>   | <div>described by the block above <math>\hat{A}^{\prime} 0.215 \hat{A} \leq 17.8</math></div>                                                                                                                 |
| <div>ctx_dose_med</div> <div>GENE REAL <math>\hat{A}^{\prime} \hat{A} \leq 0</math> 90%</div>   | <div>described by the block above <math>\hat{A}^{\prime} 0.215 \hat{A} \leq 17.8</math></div>                                                                                                                 |
| <div>ctx_frac_abund</div> <div>GENE FRACTION 0<math>\hat{A} \leq 1</math> 90%</div>             | <div>described by the block above <math>\hat{A}^{\prime} 0 \hat{A} \leq 1</math></div>                                                                                                                        |

|                                                                                                           |                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ctx_gap_core</b><br><div><div>GENE</div><div>SIGNED <math>\hat{A}^1</math></div><div>87%</div></div>   | Real-minus-decoy coupling gap for this gene. Pooled it is ~-0.012, but the per-gene IQR STRADDLES ZERO – this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. |
| <b>ctx_gap_d_dose</b><br><div><div>GENE</div><div>REAL, SIGNED</div><div>87%</div></div>                  | <i>described by the block above <math>\hat{A} \cdot -5.41</math> – <math>1.74</math></i>                                                                                                           |
| <b>ctx_gap_deconv</b><br><div><div>GENE</div><div>SIGNED <math>\hat{A}^1</math></div><div>87%</div></div> | <i>described by the block above <math>\hat{A} \cdot -0.33</math> – <math>+0.374</math></i>                                                                                                         |
| <b>ctx_gap_retention</b><br><div><div>GENE</div><div>REAL, SIGNED</div><div>83%</div></div>               | <i>described by the block above <math>\hat{A} \cdot -16.5</math> – <math>17</math></i>                                                                                                             |
| <b>ctx_measurable</b><br><div><div>GENE</div><div>BOOLEAN</div><div>98%</div></div>                       | Is the gene's coupling measurable at all given its design (positive OOF ceiling). False for 877/1549 genes.                                                                                        |
| <b>ctx_n_abund</b><br><div><div>GENE</div><div>COUNT</div><div>90%</div></div>                            | <i>described by the block above <math>\hat{A} \cdot 0</math> – <math>53</math></i>                                                                                                                 |
| <b>ctx_n_fam_abund</b><br><div><div>GENE</div><div>COUNT</div><div>90%</div></div>                        | <i>described by the block above <math>\hat{A} \cdot 0</math> – <math>34</math></i>                                                                                                                 |
| <b>ctx_n_fam_multi</b><br><div><div>GENE</div><div>COUNT</div><div>90%</div></div>                        | <i>described by the block above <math>\hat{A} \cdot 0</math> – <math>14</math></i>                                                                                                                 |

**TOP\_  $\hat{A} \cdot 8$**   
supporting – read these when the core raises a question

The gene's leading regulator and its value, by magnitude or by identity.

8 columns  $\hat{A} \cdot$  coverage **78–100%** (median 89%)  $\hat{A} \cdot$  3 share the block description

3x identifier  $\hat{A} \cdot$  2x real  $\hat{A} \cdot$  1x fraction  $\hat{A} \cdot$  1x signed  $\hat{A}^1$   $\hat{A} \cdot$  1x count

|                                                                                                                                        |                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>top_beta</b><br><div><div>GENE</div><div>REAL <math>\hat{A} \cdot 0</math></div><div>AGG FAMILY</div><div>100%</div></div>          | <i>described by the block above <math>\hat{A} \cdot 0.00096</math> – <math>1.52</math></i>                                                                                            |
| <b>top_beta_frac</b><br><div><div>GENE</div><div>FRACTION <math>\hat{A} \cdot 1</math></div><div>AGG FAMILY</div><div>100%</div></div> | <i>described by the block above <math>\hat{A} \cdot 0.076</math> – <math>1</math></i>                                                                                                 |
| <b>top_family_identity</b><br><div><div>GENE</div><div>IDENTIFIER</div><div>AGG FAMILY</div><div>87%</div></div>                       | The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything – an honest 'undefined', which the magnitude answer cannot express. |
| <b>top_family_magnitude</b><br><div><div>GENE</div><div>IDENTIFIER</div><div>AGG FAMILY</div><div>100%</div></div>                     | The family with the largest beta.                                                                                                                                                     |
| <b>top_gaining_arms</b><br><div><div>GENE</div><div>IDENTIFIER</div><div>91%</div></div>                                               | <i>described by the block above <math>\hat{A} \cdot 1,279</math> distinct</i>                                                                                                         |
| <b>top_identity</b><br><div><div>GENE</div><div>REAL <math>\hat{A} \cdot 0</math></div><div>AGG FAMILY</div><div>87%</div></div>       | The largest <b>identity</b> share among the gene's families.                                                                                                                          |

|                                                                                      |                                                                                                                                                                           |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>top_identity_gated</b>                                                            | <code>top_identity</code> recomputed over <code>identity_reliable</code> edges only â€” <b>max 1.0000, 0 genes above 1</b> (vs +740.007 ungated). Defined on 1,203 genes. |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^{+1} \in \{1\}</math></div> <div>78%</div> |                                                                                                                                                                           |
| <b>top_identity_n_reliable</b>                                                       | How many of the gene's families survive the identity gate â€” the DENOMINATOR behind <code>top_identity_gated</code> . <b>Median 2.</b>                                   |
| <div>GENE</div> <div>COUNT</div> <div>78%</div>                                      |                                                                                                                                                                           |

**REALIZED\_  $\hat{A}$  5**  
supporting â€” read these when the core raises a question

Realized (as opposed to potential) coupling for the gene â€” the raw and adjusted rho, its retention, and how many regulators it was computed over.

5 columns  $\hat{A}$  coverage **63%â€”82%** (median 81%)  $\hat{A}$  4 share the block description  
2x signed  $\hat{A}^{+1} \in \{1\}$   $\hat{A}$  1x boolean  $\hat{A}$  1x count  $\hat{A}$  1x real, signed

|                                                   |                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------|
| <b>realized_composition_explained</b>             | <i>described by the block above <math>\hat{A}</math> 490 true / 785 false</i> |
| <div>GENE</div> <div>BOOLEAN</div> <div>82%</div> |                                                                               |

|                                                 |                                                                   |
|-------------------------------------------------|-------------------------------------------------------------------|
| <b>realized_n_reg</b>                           | <i>described by the block above <math>\hat{A}</math> 1 â€” 63</i> |
| <div>GENE</div> <div>COUNT</div> <div>82%</div> |                                                                   |

|                                                        |                                                                                     |
|--------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>realized_retention</b>                              | Realized-coupling retention, $\hat{I}_{\text{adjusted}}$ / $\hat{I}_{\text{raw}}$ . |
| <div>GENE</div> <div>REAL, SIGNED</div> <div>63%</div> |                                                                                     |

|                                                                                      |                                                                            |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>realized_rho_adj</b>                                                              | <i>described by the block above <math>\hat{A}</math> -0.479 â€” +0.509</i> |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^{+1} \in \{1\}</math></div> <div>81%</div> |                                                                            |

|                                                                                      |                                                                          |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>realized_rho_raw</b>                                                              | <i>described by the block above <math>\hat{A}</math> -0.67 â€” +0.62</i> |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^{+1} \in \{1\}</math></div> <div>81%</div> |                                                                          |

**W\_  $\hat{A}$  1**  
supporting â€” read these when the core raises a question

1 columns  $\hat{A}$  coverage **100%â€”100%** (median 100%)  
1x real  $\hat{A} \leq 0$

|                                                                             |                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>w_max</b>                                                                | The MAXIMUM curated evidence weight <code>w</code> over the gene's regulators ( <code>gene_atlas : nanmax(w)</code> ), where <code>w</code> comes from the same PMID-deduped ledger that feeds <code>lit_</code> and <code>fame_</code> . |
| <div>GENE</div> <div>REAL <math>\hat{A} \leq 0</math></div> <div>100%</div> |                                                                                                                                                                                                                                           |

**LIT\_  $\hat{A}$  7**  
supporting â€” read these when the core raises a question

$\hat{A}$ â””â”” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€” read the name with suspicion.  
`eval/lit_ground_truth.py` defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded).

7 columns  $\hat{A}$  coverage **21%â€”21%** (median 21%)  $\hat{A}$  3 share the block description  
4x count  $\hat{A}$  2x boolean  $\hat{A}$  1x identifier

|                                                   |                                                                                                                                      |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>lit_agrees_identity</b>                        | Does <code>identity</code> name the same family as the <b>most-published</b> one for this gene â€” <b>True on 100 of 323 (31%)</b> . |
| <div>GENE</div> <div>BOOLEAN</div> <div>21%</div> |                                                                                                                                      |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>lit_agrees_magnitude</b><br>GENE BOOLEAN 21% | <i>described by the block above</i> $\hat{A} \cdot 96$ true / 233 false<br>  <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>                                                                                                                                                                                                                       |
| <b>lit_family</b><br>GENE IDENTIFIER 21%        | The most-published seed family for this gene by distinct low-throughput functional PMIDs.                                                                                                                                                                                                                                                                                                                       |
| <b>lit_margin</b><br>GENE COUNT 21%             | Gap between the top and second literature family, in PMIDs $\hat{A}$ ” <b>median 1.00</b> , i.e. the literature's 'winner' is usually decided by a SINGLE paper. $\hat{A}$ ” <b>gate on this before reading</b> <b>lit_agrees_*</b> : a 1-PMID margin is a coin flip dressed as ground truth.<br>  <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i> |
| <b>lit_n_lit_families</b><br>GENE COUNT 21%     | <i>described by the block above</i> $\hat{A} \cdot 2$ $\hat{A}$ ” 67<br>  <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>                                                                                                                                                                                                                          |
| <b>lit_n_pmid</b><br>GENE COUNT 21%             | Distinct low-throughput functional PMIDs behind <b>lit_family</b> . Median <b>3</b> $\hat{A}$ ” the 'ground truth' rests on a handful of papers per gene.<br>  <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>                                                                                                                                     |
| <b>lit_tier</b><br>GENE COUNT 21%               | <i>described by the block above</i> $\hat{A} \cdot 2$ $\hat{A}$ ” 4<br>  <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>                                                                                                                                                                                                                           |

**CPTAC\_  $\hat{A} \cdot 24$**   
reference  $\hat{A}$ ” context, provenance and rarely-quoted detail

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>CPTAC protein channel.</div> <div><b>24</b> columns <math>\hat{A} \cdot</math> coverage <b>42%</b><math>\hat{A}</math>”<b>89%</b> (median 66%) <math>\hat{A} \cdot</math> <b>18</b> share the block description<br/>18<math>\times</math> signed <math>\hat{A}^{\wedge}1\hat{A}</math>”1 <math>\hat{A} \cdot</math> 2<math>\times</math> boolean <math>\hat{A} \cdot</math> 2<math>\times</math> real, signed <math>\hat{A} \cdot</math> 2<math>\times</math> count</div> |                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>cptac_prosp_abund_rho_disc</b><br>GENE SIGNED $\hat{A}^{\wedge}1\hat{A}$ ”1<br>AGG ARM 72%                                                                                                                                                                                                                                                                                                                                                                                  | <i>described by the block above</i> $\hat{A} \cdot -0.546$ $\hat{A}$ ” $+0.599$<br>  <i>Defined on: genes/edges present in the CPTAC cohort</i>                                                                                                                                                                                                                                         |
| <b>cptac_prosp_abund_rho_prot</b><br>GENE SIGNED $\hat{A}^{\wedge}1\hat{A}$ ”1<br>AGG ARM 73%                                                                                                                                                                                                                                                                                                                                                                                  | <i>described by the block above</i> $\hat{A} \cdot -0.639$ $\hat{A}$ ” $+0.575$<br>  <i>Defined on: genes/edges present in the CPTAC cohort</i>                                                                                                                                                                                                                                         |
| <b>cptac_prosp_abund_rho_rna</b><br>GENE SIGNED $\hat{A}^{\wedge}1\hat{A}$ ”1<br>AGG ARM 74%                                                                                                                                                                                                                                                                                                                                                                                   | <i>described by the block above</i> $\hat{A} \cdot -0.391$ $\hat{A}$ ” $+0.582$<br>  <i>Defined on: genes/edges present in the CPTAC cohort</i>                                                                                                                                                                                                                                         |
| <b>cptac_prosp_agg_beats_abund_prot</b><br>GENE BOOLEAN 73%                                                                                                                                                                                                                                                                                                                                                                                                                    | Does the learned $\hat{I}^2$ -weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort?                                                                                                                                                                                                                                         |
| <b>cptac_prosp_agg_ret_prot</b><br>GENE REAL, SIGNED 59%                                                                                                                                                                                                                                                                                                                                                                                                                       | $\text{rho\_prot}(\text{composition-adjusted}) / \text{rho\_prot\_raw}$ $\hat{A}$ ” the ADJUSTMENT sense of retention, on the prospective cohort. $\hat{A}$ ”... <b>The denominator is ALREADY GATED</b> at <b>RHO_GATE</b> in <b>cptac_card.py:322</b> , so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with $ \text{raw}  < 0.02$ ). |
| <b>cptac_prosp_agg_rho_disc</b><br>GENE SIGNED $\hat{A}^{\wedge}1\hat{A}$ ”1<br>AGG ARM 72%                                                                                                                                                                                                                                                                                                                                                                                    | <i>described by the block above</i> $\hat{A} \cdot -0.559$ $\hat{A}$ ” $+0.502$<br>  <i>Defined on: genes/edges present in the CPTAC cohort</i>                                                                                                                                                                                                                                         |

|                                                                                                                                                                                |                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>cptac_prosp_agg_rho_disc_raw</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>72%</div></div></div>               | <div>described by the block above <math>\hat{A} \cdot -0.648 \hat{A} \epsilon'' + 0.6</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div>   |
| <div><div>cptac_prosp_agg_rho_prot</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>AGG ARM</div><div>73%</div></div></div> | <div>described by the block above <math>\hat{A} \cdot -0.639 \hat{A} \epsilon'' + 0.501</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_agg_rho_prot_raw</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>73%</div></div></div>               | <div>described by the block above <math>\hat{A} \cdot -0.654 \hat{A} \epsilon'' + 0.524</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_agg_rho_rna</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>AGG ARM</div><div>74%</div></div></div>  | <div>described by the block above <math>\hat{A} \cdot -0.465 \hat{A} \epsilon'' + 0.489</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_agg_rho_rna_raw</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>74%</div></div></div>                | <div>described by the block above <math>\hat{A} \cdot -0.546 \hat{A} \epsilon'' + 0.564</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_prosp_n_arms</div><div><div>GENE</div><div>COUNT</div><div>89%</div></div></div>                                                                               | Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61).                                                                                     |
| <div><div>cptac_t105_abund_rho_disc</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>60%</div></div></div>                  | <div>described by the block above <math>\hat{A} \cdot -0.359 \hat{A} \epsilon'' + 0.443</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_abund_rho_prot</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>60%</div></div></div>                  | <div>described by the block above <math>\hat{A} \cdot -0.368 \hat{A} \epsilon'' + 0.446</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_abund_rho_rna</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>60%</div></div></div>                   | <div>described by the block above <math>\hat{A} \cdot -0.465 \hat{A} \epsilon'' + 0.49</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div>  |
| <div><div>cptac_t105_agg_beats_abund_prot</div><div><div>GENE</div><div>BOOLEAN</div><div>60%</div></div></div>                                                                | As the prospective twin, on the TCGA-105 overlap cohort.                                                                                                              |
| <div><div>cptac_t105_agg_ret_prot</div><div><div>GENE</div><div>REAL, SIGNED</div><div>42%</div></div></div>                                                                   | As the prospective twin, TCGA-105.                                                                                                                                    |
| <div><div>cptac_t105_agg_rho_disc</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>AGG ARM</div><div>60%</div></div></div>  | <div>described by the block above <math>\hat{A} \cdot -0.412 \hat{A} \epsilon'' + 0.443</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_agg_rho_disc_raw</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>60%</div></div></div>                | <div>described by the block above <math>\hat{A} \cdot -0.334 \hat{A} \epsilon'' + 0.421</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_agg_rho_prot</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>AGG ARM</div><div>60%</div></div></div>  | <div>described by the block above <math>\hat{A} \cdot -0.46 \hat{A} \epsilon'' + 0.446</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div>  |
| <div><div>cptac_t105_agg_rho_prot_raw</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>60%</div></div></div>                | <div>described by the block above <math>\hat{A} \cdot -0.484 \hat{A} \epsilon'' + 0.446</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div> |
| <div><div>cptac_t105_agg_rho_rna</div><div><div>GENE</div><div>SIGNED <math>\hat{A}^{\prime 1} \hat{A} \epsilon   1</math></div><div>AGG ARM</div><div>60%</div></div></div>   | <div>described by the block above <math>\hat{A} \cdot -0.56 \hat{A} \epsilon'' + 0.357</math></div> <div>  Defined on: genes/edges present in the CPTAC cohort</div>  |

|                                                                         |                                                                                  |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>cptac_t105_agg_rho_rna_raw</b>                                       | <i>described by the block above</i> $-0.636 \pm 0.54$                            |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^1</math></div> <div>60%</div> | <i>Defined on: genes/edges present in the CPTAC cohort</i>                       |
| <b>cptac_t105_n_arms</b>                                                | Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. |
| <div>GENE</div> <div>COUNT</div> <div>85%</div>                         | <i>Defined on: genes/edges present in the CPTAC cohort</i>                       |

**DOMINANT\_  $\hat{A}^4$**

reference  $\hat{A}$  context, provenance and rarely-quoted detail

The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to `regulatory_handoff`. **4** columns  $\hat{A}$  coverage **77%~~83%~~** (median 82%)  $\hat{A}$  **3** share the block description  
3 $\times$  identifier  $\hat{A}$  1 $\times$  categorical

|                                                       |                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>dominant_HLY</b>                                   | <i>described by the block above</i> $\hat{A}^280$ distinct                                                         |
| <div>GENE</div> <div>IDENTIFIER</div> <div>77%</div>  |                                                                                                                    |
| <b>dominant_NAT</b>                                   | <i>described by the block above</i> $\hat{A}^309$ distinct                                                         |
| <div>GENE</div> <div>IDENTIFIER</div> <div>81%</div>  |                                                                                                                    |
| <b>dominant_TUM</b>                                   | The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). |
| <div>GENE</div> <div>IDENTIFIER</div> <div>82%</div>  |                                                                                                                    |
| <b>dominant_edge_shift_class</b>                      | <i>described by the block above</i> $\hat{A}^10$ levels                                                            |
| <div>GENE</div> <div>CATEGORICAL</div> <div>83%</div> |                                                                                                                    |

**FRAC\_  $\hat{A}^3$**

reference  $\hat{A}$  context, provenance and rarely-quoted detail

A fraction  $\hat{A}$  read the suffix for the numerator and `domain` for the denominator's rows. **3** columns  $\hat{A}$  coverage **60%~~100%~~** (median 82%)  $\hat{A}$  **3** share the block description  
3 $\times$  fraction  $\hat{A}$ 1

|                                                                            |                                                        |
|----------------------------------------------------------------------------|--------------------------------------------------------|
| <b>frac_edges_realized</b>                                                 | <i>described by the block above</i> $\hat{A}^0 \pm 1$  |
| <div>GENE</div> <div>FRACTION <math>\hat{A}^1</math></div> <div>82%</div>  |                                                        |
| <b>frac_gain_true_acquired</b>                                             | <i>described by the block above</i> $\hat{A}^0 \pm 1$  |
| <div>GENE</div> <div>FRACTION <math>\hat{A}^1</math></div> <div>60%</div>  | <i>Defined on: genes with a NAT-TUM gene-level leg</i> |
| <b>frac_identified</b>                                                     | <i>described by the block above</i> $\hat{A}^0 \pm 1$  |
| <div>GENE</div> <div>FRACTION <math>\hat{A}^1</math></div> <div>100%</div> |                                                        |

**STATIC\_  $\hat{A}^1$**

reference  $\hat{A}$  context, provenance and rarely-quoted detail

**1** columns  $\hat{A}$  coverage **87%~~87%~~** (median 87%)  
1 $\times$  identifier

|                                                      |                                                                                            |
|------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>static_owner_family</b>                           | The static (dose-based) owner of the gene, for comparison against the realization owner.   |
| <div>GENE</div> <div>IDENTIFIER</div> <div>87%</div> | <i>Defined on: MULTI-FAMILY genes only (an owner needs &gt;1 family to choose between)</i> |

TOTAL\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 100%“100% (median 100%)  
1x real 0

total\_pressure

Sum over the gene's regulators.

GENE

REAL 0

100%

OWNER\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 41%“41% (median 41%)  
1x boolean

owner\_agrees

Whether the realization owner and the static owner agree. Defined only where both exist.  

GENE

BOOLEAN

41%

Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

MEDIAN\_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 100%“100% (median 100%)  
1x real 0

median\_retention

A per-row median of a lower-rung quantity; check `agg_of`.

GENE

REAL 0

100%

ARMRES\_ 7

reference “ context, provenance and rarely-quoted detail

â IS THIS GENE MODELABLE AT ARM RESOLUTION “ the gene-level roll-up of arm identifiability  
(`card_ladders.gene_arm_resolution`, 2026-08-19).  
7 columns coverage 18%“18% (median 18%) 4 share the block description  
2x signed 1 1 2x count 1x categorical 1x fraction 1 1x real 0

armres\_best\_drho

described by the block above -0.206 “ +0.0421

GENE

SIGNED 1 1

18%

armres\_class

`arm_modelable` (every multi-arm cell resolvable, 68 genes) partial (37) family\_only (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined.

GENE

CATEGORICAL

18%

armres\_frac\_resolvable

Fraction of this gene's multi-arm cells where the arms are separable.

GENE

FRACTION 1

18%

armres\_med\_drho

Median `oof_drho` over the gene's multi-arm cells “ negative means arm resolution predicts repression better out of fold.

GENE

SIGNED 1 1

18%

armres\_med\_sep\_z

described by the block above 0.0536 “ 18.5

GENE

REAL 0

18%

|                                                 |                                                                          |
|-------------------------------------------------|--------------------------------------------------------------------------|
| <b>armres_n_arms_split</b>                      | <i>described by the block above</i> $\hat{A} \cdot 2$ $\hat{a} \in$ " 40 |
| <div>GENE</div> <div>COUNT</div> <div>18%</div> |                                                                          |
| <b>armres_n_multi_cells</b>                     | <i>described by the block above</i> $\hat{A} \cdot 1$ $\hat{a} \in$ " 14 |
| <div>GENE</div> <div>COUNT</div> <div>18%</div> |                                                                          |

IDENTITY\_  $\hat{A} \cdot 1$

reference  $\hat{a} \in$  " context, provenance and rarely-quoted detail

|                                                                                                                                               |                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>1 columns <math>\hat{A} \cdot</math> coverage <b>87%<del><math>\hat{a} \in</math></del>"87%</b> (median 87%)</div> <div>1x boolean</div> |                                                                                                                                                                                    |
| <b>identity_eq_magnitude</b>                                                                                                                  | Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at $n\_fam=1$ by construction; 76% at $n\_fam \geq 2$ . Read it only on multi-family genes. |
| <div>GENE</div> <div>BOOLEAN</div> <div>AGG FAMILY</div>                                                                                      |                                                                                                                                                                                    |
| <div>87%</div>                                                                                                                                |                                                                                                                                                                                    |

HEUR\_  $\hat{A} \cdot 5$

reference  $\hat{a} \in$  " context, provenance and rarely-quoted detail

|                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>5 columns <math>\hat{A} \cdot</math> coverage <b>89%<del><math>\hat{a} \in</math></del>"91%</b> (median 91%)</div> <div>2x signed <math>\hat{a}^{'1} \hat{a} \in</math>   1 <math>\hat{A} \cdot</math> 1x count <math>\hat{A} \cdot</math> 1x boolean <math>\hat{A} \cdot</math> 1x categorical</div> |                                                                                                                                                                                                                                                            |
| <b>heur_delta_tumor_nat</b>                                                                                                                                                                                                                                                                                | Tumour-minus-NAT change in the heuristic pressure correlation.                                                                                                                                                                                             |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^{'1} \hat{A} \in</math>   1</div> <div>89%</div>                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                            |
| <b>heur_n_regulators</b>                                                                                                                                                                                                                                                                                   | $\hat{a} \hat{s} \hat{a} \hat{s}$ <b>A DIFFERENT PIPELINE'S COUNT <math>\hat{a} \in</math> " not the model's.</b> It is <code>edges.groupby('gene')['miRNA'].nunique()</code> inside <code>analyses/misc/mirna_comovement.py</code> , i.e.                 |
| <div>GENE</div> <div>COUNT</div> <div>91%</div>                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                            |
| <b>heur_net_repressed_tumor</b>                                                                                                                                                                                                                                                                            | Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes.                                                                                                                                                          |
| <div>GENE</div> <div>BOOLEAN</div> <div>91%</div>                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                            |
| <b>heur_repression_class</b>                                                                                                                                                                                                                                                                               | Cross-state repression class from the RETIRED heuristic lane ( <code>never_repressed</code> / <code>lost_repression</code> / <code>gained_repression</code> / <code>constitutive_repressed</code> ). 867 of 1,409 genes are <code>never_repressed</code> . |
| <div>GENE</div> <div>CATEGORICAL</div> <div>91%</div>                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                            |
| <b>heur_rho_pressure_tumor</b>                                                                                                                                                                                                                                                                             | Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour.                                                                                                                                                                            |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^{'1} \hat{A} \in</math>   1</div> <div>91%</div>                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                            |

GREAL\_  $\hat{A} \cdot 10$

reference  $\hat{a} \in$  " context, provenance and rarely-quoted detail

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Realization ladder for the GENE <math>\hat{a} \in</math> " the mirror of <code>real_1</code>: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). <math>\hat{a} \in</math> ... Cross-rung control: the gene-side totals reproduce the arm-side exactly.</div> <div>10 columns <math>\hat{A} \cdot</math> coverage <b>29%<del><math>\hat{a} \in</math></del>"83%</b> (median 83%) <math>\hat{A} \cdot</math> 8 share the block description</div> <div>7x count <math>\hat{A} \cdot</math> 2x signed <math>\hat{a}^{'1} \hat{a} \in</math>   1 <math>\hat{A} \cdot</math> 1x fraction 0<math>\hat{a} \in</math> "1</div> |                                                                                                                                                                                                         |
| <b>greal_best_coupling</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>described by the block above</i> $\hat{A} \cdot -0.555$ $\hat{a} \in$ " +0.298                                                                                                                       |
| <div>GENE</div> <div>SIGNED <math>\hat{A}^{'1} \hat{A} \in</math>   1</div> <div>83%</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Defined on: genes with <math>\hat{a} \% \geq 1</math> calibrated-coupling-scored regulator (1,260) <math>\hat{a} \in</math> " the GENE-side ladder; rates gated <math>n \hat{a} \% \geq 3</math></i> |

|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>greal_frac_c10</div> <div><div>GENE</div><div>FRACTION 0</div><div>29%</div></div>   | <div>Gene-rung twin of real_frac_c10 defined on 446 genes against greal_n_c10's 1,289; gated-out genes have a median greal_n_scored of 1. Exact on 100%.</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                    |
| <div>greal_med_coupling</div> <div><div>GENE</div><div>SIGNED 1</div><div>83%</div></div> | <div>described by the block above -0.503 +0.298</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                             |
| <div>greal_n_c05</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>           | <div>described by the block above 0 41</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                                      |
| <div>greal_n_c10</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>           | <div>described by the block above 0 33</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                                      |
| <div>greal_n_c15</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>           | <div>described by the block above 0 29</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                                      |
| <div>greal_n_c20</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>           | <div>described by the block above 0 24</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                                      |
| <div>greal_n_c30</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>           | <div>described by the block above 0 9</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                                       |
| <div>greal_n_scored</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>        | <div>Genes/edges scored for realized coupling (median 2, max 59) the DENOMINATOR of every greal_frac_*. The greal_n_c05   n_c30 ladder is properly nested: <b>0 monotonicity violations and 0 rows where a threshold count exceeds n_scored</b> (verified).</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div> |
| <div>greal_n_sig</div> <div><div>GENE</div><div>COUNT</div><div>83%</div></div>           | <div>described by the block above 0 28</div> <div>Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated 3</div>                                                                                                                                                                                                                      |

ACQUIRED\_ 2

reference context, provenance and rarely-quoted detail

Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).

2 columns coverage 91%91% (median 91%) 2 share the block description

2x real, signed

|                                                                                             |                                                    |
|---------------------------------------------------------------------------------------------|----------------------------------------------------|
| <div>acquired_vs_gtex</div> <div><div>GENE</div><div>REAL, SIGNED</div><div>91%</div></div> | <div>described by the block above -11.6 7.56</div> |
| <div>acquired_vs_nat</div> <div><div>GENE</div><div>REAL, SIGNED</div><div>91%</div></div>  | <div>described by the block above -4.24 2.83</div> |

PRESSURE\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 91%91% (median 91%)  
1x real 0

pressure\_tumor Aggregate incoming pressure on the gene in the named state.

GENE REAL 0 91%

REALIZATION\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 42%42% (median 42%)  
1x identifier

realization\_owner Within-patient realization quantities " the score, its identity allocation, and who owns it.

GENE IDENTIFIER 42%

| Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

TCGA\_ 4

reference " context, provenance and rarely-quoted detail

The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.  
4 columns coverage 89%89% (median 89%) 3 share the block description  
3x signed 1 1 1x count

tcga\_abund\_rho\_rna described by the block above -0.36 " +0.366

GENE SIGNED 1 1

AGG ARM 89%

tcga\_agg\_rho\_rna Gene-level  $\hat{l}^2$ -weighted aggregate coupling to mRNA in TCGA (median  $\hat{a}^{*0.05}$ ) against tcga\_abund\_rho\_rna's  $\hat{a}^{*0.02}$  " the abundance-sum reference.

GENE SIGNED 1 1

AGG ARM 89%

tcga\_agg\_rho\_rna\_raw described by the block above -0.545 " +0.571

GENE SIGNED 1 1 89%

tcga\_n\_arms described by the block above 1 " 62

GENE COUNT 89%

DISC\_ 2

reference " context, provenance and rarely-quoted detail

2 columns coverage 3%100% (median 51%)  
1x categorical 1x count

disc\_gold\_families Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed).

GENE CATEGORICAL 3%

disc\_n\_gold\_edges How many of the 157 discovery-queue edges name this gene (max 4).

GENE COUNT 100%

MAX\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x fraction 1

max\_beta\_frac\_sd

Maximum over the row's constituent units.

GENE

FRACTION 1

100%

REGULATORY\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x boolean

regulatory\_handoff

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour.

GENE

BOOLEAN

100%

MEAN\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 82%82% (median 82%)

1x real, signed

mean\_dTarget

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises.

GENE

REAL, SIGNED

82%

CONCENTRATION\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 53%53% (median 53%)

1x fraction 1

concentration\_adj

$(\text{concentration} \wedge 1/n_{fam}) / (1 \wedge 1/n_{fam}) \wedge [0,1]$  how concentrated the gene is RELATIVE to the most diffuse its own design allows.

GENE

FRACTION 1

53%

FAM\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 81%81% (median 81%)

1x signed 1

fam\_pooled\_rho\_adj

Seed-family structure the arm sits in " membership, affinity and dominance within it.

GENE

SIGNED 11

81%

gene\_family\_card.tsv

# gene\_family

One row per one (seed family, gene) pair. Keyed [gene, family] it cannot express a property of the family itself. That is the seed\_family card's job.

61 columns   5,117 rows   12 blocks   median coverage 97%

## Contents 12 blocks, most important first

1. (bare) 8 CORE

2. n\_3 SUPPORTING

3. ctx\_16 SUPPORTING

4. beta\_7 SUPPORTING

5. identity\_4 REFERENCE

6. comp\_16 REFERENCE
7. w\_1 REFERENCE

8. pip\_2 REFERENCE

9. m\_1 REFERENCE

10. composition\_1 REFERENCE

11. prior\_1 REFERENCE

12. retention\_1 REFERENCE

### (BARE) 8

core the columns findings rest on

8 columns coverage 95%100% (median 100%)  
3x identifier 3x real 0 1x boolean 1x real, signed

|                                                                                       |                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>arms</div> <div><div>FAMILY</div><div>IDENTIFIER</div><div>100%</div></div>      | The arms collapsed into this (gene, family) cell, <b>semicolon-separated</b> ( hsa-miR-17-5p; hsa-miR-20a-5p ).                                                                 |
| <div>beta</div> <div><div>FAMILY</div><div>REAL 0</div><div>100%</div></div>          | FAMILY-rung coupling coefficient readouts.run(level='family') , the same-seed collapse APPLIED.                                                                                 |
| <div>family</div> <div><div>KEY</div><div>IDENTIFIER</div><div>100%</div></div>       | KEY seed family of the regulator, as used by the family-rung fit.                                                                                                               |
| <div>gene</div> <div><div>KEY</div><div>IDENTIFIER</div><div>100%</div></div>         | KEY HGNC gene symbol.                                                                                                                                                           |
| <div>identified</div> <div><div>FAMILY</div><div>BOOLEAN</div><div>100%</div></div>   | z  > 2 on the UNCALIBRATED SD 27.1% of cells. cal_identified is the honest version.                                                                                             |
| <div>identity</div> <div><div>FAMILY</div><div>REAL, SIGNED</div><div>95%</div></div> | Shapley/LMG identity at the FAMILY rung same estimator as the edge card's identity , different unit.                                                                            |
| <div>retention</div> <div><div>FAMILY</div><div>REAL 0</div><div>100%</div></div>     | beta_deconv / beta_core at the family grain the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. |
| <div>z</div> <div><div>FAMILY</div><div>REAL 0</div><div>100%</div></div>             | beta / beta_sd for the family cell, on the UNCALIBRATED posterior SD.                                                                                                           |

N\_ 3

supporting read these when the core raises a question

3 columns coverage 100%100% (median 100%)  
2x count 1x constant

n\_arms

FAMILY

COUNT

100%

How many arms of THIS family sit in THIS gene's design. FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's n\_arms exactly a clean cross-rung consistency check.

n\_fam

GENE

COUNT

100%

How many families compete at this gene. beta is mathematically INERT at 1.

n\_samples

GENE

CONSTANT

100%

Samples the family cell was fit on CONSTANT 1,040 across the card.

CTX\_ 16

supporting read these when the core raises a question

Gene-design context from the decoy/atlas layer ( learned/analyses/gene\_atlas.py via card\_context ): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.  
  
16 columns coverage 81%100% (median 96%) 16 share the block description  
4x signed 1 3x real 0 3x real, signed 3x count 1x categorical 1x fraction 01 1x boolean

ctx\_apriori\_class

GENE

CATEGORICAL

100%

described by the block above 4 levels

ctx\_ceiling

GENE

SIGNED 11

99%

described by the block above -0.022 +0.44

ctx\_collapse

GENE

REAL 0

96%

described by the block above 1 5

ctx\_d\_collin

GENE

SIGNED 11

81%

described by the block above -0.699 +0.495  
Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)

ctx\_d\_fam

GENE

REAL, SIGNED

95%

described by the block above -2 25

ctx\_dose\_max

GENE

REAL 0

96%

described by the block above 0.215 17.8

ctx\_dose\_med

GENE

REAL 0

96%

described by the block above 0.215 17.8

ctx\_frac\_abund

GENE

FRACTION 01

96%

described by the block above 0 1

ctx\_gap\_core

GENE

SIGNED 11

95%

described by the block above -0.476 +0.541

ctx\_gap\_d\_dose

GENE

REAL, SIGNED

95%

described by the block above -5.41 1.74

|                   |                                                                                     |     |
|-------------------|-------------------------------------------------------------------------------------|-----|
| ctx_gap_deconv    | described by the block above $\hat{\mu} = -0.33 \pm 0.374$                          |     |
| GENE              | SIGNED $\hat{\mu} \in [-1, 1]$                                                      | 95% |
| ctx_gap_retention | described by the block above $\hat{\mu} = -16.5 \pm 17$                             |     |
| GENE              | REAL, SIGNED                                                                        | 92% |
| ctx_measurable    | described by the block above $\hat{\mu} = 3,652 \text{ true} / 1,432 \text{ false}$ |     |
| GENE              | BOOLEAN                                                                             | 99% |
| ctx_n_abund       | described by the block above $\hat{\mu} = 0 \pm 53$                                 |     |
| GENE              | COUNT                                                                               | 96% |
| ctx_n_fam_abund   | described by the block above $\hat{\mu} = 0 \pm 34$                                 |     |
| GENE              | COUNT                                                                               | 96% |
| ctx_n_fam_multi   | described by the block above $\hat{\mu} = 0 \pm 14$                                 |     |
| GENE              | COUNT                                                                               | 96% |

**BETA\_  $\hat{\mu} = 7$**   
supporting  $\hat{\mu}$  read these when the core raises a question

Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta.

**7** columns  $\hat{\mu}$  coverage **100%** (median 100%)  $\hat{\mu} = 7$  share the block description  
4x fraction  $\hat{\mu} \in [-1, 1]$  3x real  $\hat{\mu} \in \mathbb{R}$

|                  |                                                                            |      |
|------------------|----------------------------------------------------------------------------|------|
| beta_deconv      | described by the block above $\hat{\mu} \cdot 0.00114 \hat{\sigma}'' 1.52$ |      |
| FAMILY           | REAL $\hat{\mu} \in \mathbb{R}$                                            | 100% |
| beta_frac        | described by the block above $\hat{\mu} \cdot 0.00165 \hat{\sigma}'' 1$    |      |
| FAMILY           | FRACTION $\hat{\mu} \in [-1, 1]$                                           | 100% |
| beta_frac_abs    | described by the block above $\hat{\mu} \cdot 0.00137 \hat{\sigma}'' 1$    |      |
| FAMILY           | FRACTION $\hat{\mu} \in [-1, 1]$                                           | 100% |
| beta_frac_deconv | described by the block above $\hat{\mu} \cdot 0.00253 \hat{\sigma}'' 1$    |      |
| FAMILY           | FRACTION $\hat{\mu} \in [-1, 1]$                                           | 100% |
| beta_frac_sd     | described by the block above $\hat{\mu} \cdot 0 \hat{\sigma}'' 0.292$      |      |
| FAMILY           | FRACTION $\hat{\mu} \in [-1, 1]$                                           | 100% |
| beta_sd          | described by the block above $\hat{\mu} \cdot 0.00082 \hat{\sigma}'' 3.21$ |      |
| FAMILY           | REAL $\hat{\mu} \in \mathbb{R}$                                            | 100% |
| beta_sd_deconv   | described by the block above $\hat{\mu} \cdot 0.00095 \hat{\sigma}'' 3.21$ |      |
| FAMILY           | REAL $\hat{\mu} \in \mathbb{R}$                                            | 100% |

IDENTITY\_ 4

reference “ context, provenance and rarely-quoted detail

|                                                                                     |                                                                                                                                                                                 |     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4 columns coverage 95%97% (median 95%)<br>2x fraction 01 1x real, signed 1x boolean |                                                                                                                                                                                 |     |
| identity_abs                                                                        | identity  / Î identity  per gene “ the BOUNDED companion to identity , always in [0,1] (verified: 0 non-NaN values outside it on either card).                                  |     |
| FAMILY                                                                              | FRACTION 01                                                                                                                                                                     | 95% |
| identity_coherence                                                                  | 1 / Î identity  per GENE, in (0,1] “ identity's own sign coherence. 1.0 when nothing cancels, “ 0 as suppressor cancellation grows.                                             |     |
| FAMILY                                                                              | FRACTION 01                                                                                                                                                                     | 95% |
| identity_deconv                                                                     | Shapley/LMG attribution on R^2 with bagged-NNLS weights “ WHO regulates the gene, collinearity-fair.                                                                            |     |
| FAMILY                                                                              | REAL, SIGNED                                                                                                                                                                    | 97% |
| identity_reliable                                                                   | identity_coherence “ 0.5 AND  identity  “ 1 . Admits 93.3% of edge rows and 92.6% of gene_family rows, and excludes every row whose  identity  exceeds 1 (119/119 and 118/118). |     |
| FAMILY                                                                              | BOOLEAN                                                                                                                                                                         | 95% |

COMP\_ 16

reference “ context, provenance and rarely-quoted detail

|                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Cell-composition drivers of this gene's coupling ( compartment_* ): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.<br>16 columns coverage 66%97% (median 81%) 7 share the block description<br>6x categorical 6x signed 11 3x real, signed 1x real “ 0 |                                                                                                                                                                                                                                |     |
| comp_cptac_prot_driver                                                                                                                                                                                                                                                                                      | described by the block above 10 levels<br>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)                                                                                                   |     |
| GENE                                                                                                                                                                                                                                                                                                        | CATEGORICAL                                                                                                                                                                                                                    | 66% |
| comp_cptac_prot_driver_ret                                                                                                                                                                                                                                                                                  | 1 “ comp_cptac_prot_driver_share , exactly. Median +0.398 raw, +0.469 gated to  _raw  “ 0.10 “ quote the gated value. See the mRNA twin.<br>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101) |     |
| GENE                                                                                                                                                                                                                                                                                                        | REAL, SIGNED                                                                                                                                                                                                                   | 66% |
| comp_cptac_prot_driver_share                                                                                                                                                                                                                                                                                | Protein-layer twin of comp_tcga_mrna_driver_share . Same exact complementarity with _driver_ret (8.9e-16 on 918 rows).                                                                                                         |     |
| GENE                                                                                                                                                                                                                                                                                                        | REAL, SIGNED                                                                                                                                                                                                                   | 66% |
| comp_cptac_prot_rho_raw                                                                                                                                                                                                                                                                                     | Unadjusted aggregate PROTEIN coupling “ denominator of comp_cptac_prot_driver_ret .                                                                                                                                            |     |
| GENE                                                                                                                                                                                                                                                                                                        | SIGNED 11                                                                                                                                                                                                                      | 81% |
| comp_cptac_prot_top_budget                                                                                                                                                                                                                                                                                  | described by the block above 10 levels<br>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)                                                                                                   |     |
| GENE                                                                                                                                                                                                                                                                                                        | CATEGORICAL                                                                                                                                                                                                                    | 81% |
| comp_cptac_prot_top_budget_r                                                                                                                                                                                                                                                                                | described by the block above 10 levels<br>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)                                                                                                   |     |
| GENE                                                                                                                                                                                                                                                                                                        | SIGNED 11                                                                                                                                                                                                                      | 81% |
| comp_cptac_prot_top_target                                                                                                                                                                                                                                                                                  | described by the block above 10 levels<br>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)                                                                                                   |     |
| GENE                                                                                                                                                                                                                                                                                                        | CATEGORICAL                                                                                                                                                                                                                    | 81% |

|                              |                                                                                                                                                                                                                                          |     |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| comp_cptac_prot_top_target_r | described by the block above $\hat{A} \cdot -0.745 \hat{A} \hat{A}'' +0.842$<br>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)                                                                       |     |
| GENE                         | SIGNED $\hat{A}^{'1}\hat{A}\hat{A}''1$                                                                                                                                                                                                   | 81% |
| comp_tcga_mrna_driver        | WHICH compartment drives the gene's mRNA coupling $\hat{A} \hat{A}''$ the one whose removal moves it most ( CAFs 279 $\hat{A} \cdot$ purity 174 $\hat{A} \cdot$ prolif 148 $\hat{A} \cdot$ Endothelial 122 $\hat{A} \cdot$ Myeloid 109). |     |
| GENE                         | CATEGORICAL                                                                                                                                                                                                                              | 81% |
| comp_tcga_mrna_driver_ret    | 1 $\hat{A}^{'1}$ comp_tcga_mrna_driver_share , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment.                                                                   |     |
| GENE                         | REAL, SIGNED                                                                                                                                                                                                                             | 81% |
| comp_tcga_mrna_driver_share  | Share of the gene's mRNA coupling attributable to the single compartment that most changes it $\hat{A} \hat{A}''$ axiom 8's composition GATE, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10.  |     |
| GENE                         | REAL $\hat{A} \hat{A}'' 0$                                                                                                                                                                                                               | 81% |
| comp_tcga_mrna_rho_raw       | The gene's UNADJUSTED aggregate mRNA coupling $\hat{A} \hat{A}''$ the DENOMINATOR of comp_tcga_mrna_driver_ret .                                                                                                                         |     |
| GENE                         | SIGNED $\hat{A}^{'1}\hat{A}\hat{A}''1$                                                                                                                                                                                                   | 97% |
| comp_tcga_mrna_top_budget    | The compartment the gene's miRNA BUDGET loads on most $\hat{A} \hat{A}''$ a property of the REGULATORS.                                                                                                                                  |     |
| GENE                         | CATEGORICAL                                                                                                                                                                                                                              | 97% |
| comp_tcga_mrna_top_budget_r  | described by the block above $\hat{A} \cdot -0.561 \hat{A} \hat{A}'' +0.53$                                                                                                                                                              |     |
| GENE                         | SIGNED $\hat{A}^{'1}\hat{A}\hat{A}''1$                                                                                                                                                                                                   | 97% |
| comp_tcga_mrna_top_target    | The compartment the TARGET GENE's expression loads on most.                                                                                                                                                                              |     |
| GENE                         | CATEGORICAL                                                                                                                                                                                                                              | 97% |
| comp_tcga_mrna_top_target_r  | described by the block above $\hat{A} \cdot -0.691 \hat{A} \hat{A}'' +0.887$                                                                                                                                                             |     |
| GENE                         | SIGNED $\hat{A}^{'1}\hat{A}\hat{A}''1$                                                                                                                                                                                                   | 97% |

W\_  $\hat{A} \cdot$  1

reference  $\hat{A} \hat{A}''$  context, provenance and rarely-quoted detail

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| 1 columns $\hat{A} \cdot$ coverage 100% $\hat{A} \hat{A}''$ 100% (median 100%)<br>1x real $\hat{A} \hat{A}'' 0$ |
|-----------------------------------------------------------------------------------------------------------------|

|       |                                                                                                             |      |
|-------|-------------------------------------------------------------------------------------------------------------|------|
| w_max | Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card. |      |
| GENE  | REAL $\hat{A} \hat{A}'' 0$                                                                                  | 100% |

PIP\_  $\hat{A} \cdot$  2

reference  $\hat{A} \hat{A}''$  context, provenance and rarely-quoted detail

|                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 columns $\hat{A} \cdot$ coverage 100% $\hat{A} \hat{A}''$ 100% (median 100%)<br>1x constant $\hat{A} \cdot$ 1x fraction 0 $\hat{A} \hat{A}''$ 1 |
|---------------------------------------------------------------------------------------------------------------------------------------------------|

|           |                                                                                      |      |
|-----------|--------------------------------------------------------------------------------------|------|
| pip_dense | $\hat{A} \hat{A}''$ CONSTANT by construction ( $\hat{A} \hat{A}'' 1$ dense readout). |      |
| FAMILY    | CONSTANT                                                                             | 100% |

|               |                                                                                                                              |  |
|---------------|------------------------------------------------------------------------------------------------------------------------------|--|
| pip_discovery | Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout. |  |
| FAMILY        | FRACTION 0 $\hat{A} \hat{A}''$ 1                                                                                             |  |
| 100%          |                                                                                                                              |  |

M\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x real 0

m\_nnls Bagged-NNLS weight for this (gene, family) cell.

FAMILY

REAL 0

100%

COMPOSITION\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x categorical

composition\_class Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block).

FAMILY

CATEGORICAL

100%

PRIOR\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x fraction 01

prior\_pi The evidence prior fed to the discovery readout.

FAMILY

FRACTION 01

100%

RETENTION\_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x boolean

retention\_reliable Fraction of an effect surviving covariate adjustment (adjusted / raw).

FAMILY

BOOLEAN

100%

**arm**

**296** columns   **2,450** rows   **43** blocks   median coverage **30%**

1. adm\_ 4 Â· CORE
2. arm\_ 7 Â· CORE
3. seed\_ 1 Â· CORE
4. site\_ 12 Â· SUPPORTING
5. arb\_ 16 Â· SUPPORTING
6. healthy\_ 3 Â· SUPPORTING
7. (bare) 4 Â· SUPPORTING
8. fame\_ 28 Â· SUPPORTING
9. ago\_ 4 Â· SUPPORTING
10. hly\_ 7 Â· SUPPORTING
11. dose\_ 3 Â· SUPPORTING
12. abund\_ 5 Â· SUPPORTING
13. attr\_ 14 Â· REFERENCE
14. ctx\_ 2 Â· REFERENCE
15. grank\_ 3 Â· REFERENCE
16. real\_ 11 Â· REFERENCE
17. ts\_ 10 Â· REFERENCE
18. isoc\_ 5 Â· REFERENCE
19. sdsz\_ 4 Â· REFERENCE
20. chim\_ 5 Â· REFERENCE
21. fst\_ 16 Â· REFERENCE
22. sub\_ 6 Â· REFERENCE
23. seq\_ 4 Â· REFERENCE
24. cur\_ 2 Â· REFERENCE
25. cov\_ 22 Â· REFERENCE
26. iso\_ 5 Â· REFERENCE
27. field\_ 5 Â· REFERENCE
28. famrole\_ 6 Â· REFERENCE
29. surv\_ 8 Â· REFERENCE
30. aid\_ 4 Â· REFERENCE
31. bc\_ 11 Â· REFERENCE
32. wshift\_ 6 Â· REFERENCE
33. tier\_ 5 Â· REFERENCE
34. disc\_ 2 Â· REFERENCE
35. cnv\_ 9 Â· REFERENCE
36. gctx\_ 16 Â· REFERENCE
37. surrogate\_ 2 Â· REFERENCE
38. cnvc\_ 5 Â· REFERENCE
39. foc\_ 7 Â· REFERENCE
40. model\_ 1 Â· REFERENCE
41. comv\_ 2 Â· REFERENCE
42. cload\_ 3 Â· REFERENCE
43. hx\_ 1 Â· REFERENCE

4 columns → coverage 14% → 29% (median 29%)  
3x count → 1x fraction → 1

|     |                              |     |
|-----|------------------------------|-----|
| ARM | FRACTION $\theta \hat{=}$ "1 | 14% |
|-----|------------------------------|-----|

|     |       |     |
|-----|-------|-----|
| ARM | COUNT | 29% |
|-----|-------|-----|

| ARM   | COUNT | PERCENT |
|-------|-------|---------|
| ARM 1 | 1     | 29%     |

|                                                                                |                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>adm_n_with_site</b> <div><div>ARM</div><div>COUNT</div><div>29%</div></div> | Edges of this arm whose target carries a seed site â€” <b>89.9% of edges (4,436/4,937)</b> , so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none. <div><div>Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% Â· expressed 63.9% Â· admissible 60.4%)</div></div> |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**ARM\_ 7**  
core â€” the columns findings rest on

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <b>7</b> columns Â· coverage <b>15%â€”20%</b> (median 20%)<br>4x real, signed Â· 2x real â‰¥ 0 Â· 1x percent |
|--------------------------------------------------------------------------------------------------------------|

|                                                                                           |                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>arm_iqr</b> <div><div>ARM</div><div>REAL â‰¥ 0</div><div>20%</div></div>               | Interquartile range of the arm's tumour RPM â€” its DYNAMIC RANGE.                                                                                                                                                                                 |
| <b>arm_lfc_HLY_NAT_raw</b> <div><div>ARM</div><div>REAL, SIGNED</div><div>15%</div></div> | log-FC GTEEx-healthy â†’ TCGA-NAT, raw.                                                                                                                                                                                                            |
| <b>arm_lfc_HLY_TUM_QN</b> <div><div>ARM</div><div>REAL, SIGNED</div><div>16%</div></div>  | log-FC GTEEx-healthy â†’ TCGA-tumour, on quantile-normalised values.                                                                                                                                                                               |
| <b>arm_lfc_HLY_TUM_raw</b> <div><div>ARM</div><div>REAL, SIGNED</div><div>15%</div></div> | As <code>arm_lfc_HLY_TUM_QN</code> but against the RAW GTEEx median.                                                                                                                                                                               |
| <b>arm_lfc_NAT_TUM</b> <div><div>ARM</div><div>REAL, SIGNED</div><div>20%</div></div>     | log-FC of the arm's median abundance, matched NAT â†’ tumour.                                                                                                                                                                                      |
| <b>arm_med_rpm</b> <div><div>ARM</div><div>REAL â‰¥ 0</div><div>20%</div></div>           | The arm's MEDIAN RPM across tumour samples â€” its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. <div><div>Defined on: edges present in the progression panel (arm expressed + state measured)</div></div> |
| <b>arm_pct_above_floor</b> <div><div>ARM</div><div>PERCENT</div><div>20%</div></div>      | â€” <b>THE NAME READS BACKWARDS</b> . Computed as <code>100Â·mean(x &gt; FLOOR)</code> â€” the percent of samples ABOVE the detection floor, i.e.                                                                                                  |

**SEED\_ 1**  
core â€” the columns findings rest on

|                                                                                |
|--------------------------------------------------------------------------------|
| <b>1</b> columns Â· coverage <b>100%â€”100%</b> (median 100%)<br>1x identifier |
|--------------------------------------------------------------------------------|

|                                                                                  |                                                     |
|----------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>seed_family</b> <div><div>ARM</div><div>IDENTIFIER</div><div>100%</div></div> | KEY â€” the seed family itself, one row per family. |
|----------------------------------------------------------------------------------|-----------------------------------------------------|

**SITE\_ 12**  
supporting â€” read these when the core raises a question

|                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.<br><b>12</b> columns Â· coverage <b>29%â€”29%</b> (median 29%) Â· <b>7</b> share the block description<br>7x count Â· 2x fraction 0â€”1 Â· 2x real, signed Â· 1x real â‰¥ 0 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                        |                                                                                                               |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>site_frac_8mer</b> <div><div>ARM</div><div>FRACTION 0â€”1</div><div>29%</div></div> | 8mer share of the arm's <b>CANONICAL</b> sites ( <code>site_n_8mer / site_n_canonical</code> ; median 0.150). |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|

|                                                                                                                  |                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>site_frac_canonical</div> <div><div>ARM</div><div>FRACTION 0Å€"1</div><div>29%</div></div>                  | Canonical share of the arm's <b>TOTAL</b> sites ( <code>site_n_canonical / site_n_total</code> ; median 0.054) Å€" a different denominator from <code>site_frac_8mer</code> .                                                                                                                                         |
| <div>site_n_6mer</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>                | <i>described by the block above</i> Å·115 Å€" 6559<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                         |
| <div>site_n_7mer</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>                | <i>described by the block above</i> Å·9 Å€" 2017<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                           |
| <div>site_n_8mer</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>                | <i>described by the block above</i> Å·0 Å€" 363<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                            |
| <div>site_n_canonical</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>           | <i>described by the block above</i> Å·9 Å€" 2295<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                           |
| <div>site_n_genes_canonical</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>     | <i>described by the block above</i> Å·9 Å€" 796<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                            |
| <div>site_n_noncanon</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>            | <i>described by the block above</i> Å·78 Å€" 36336<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                         |
| <div>site_n_total</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>29%</div></div>               | <i>described by the block above</i> Å·451 Å€" 40511<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>                                                                                                                                                        |
| <div>site_repression_med</div> <div><div>ARM</div><div>REAL, SIGNED</div><div>AGG EDGE</div><div>29%</div></div> | Median predicted repression score over the arm's sites (all negative by construction; median Å·0.316). A PREDICTION from sequence, not a measurement Å€" never cite it as evidence of observed repression.<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i> |
| <div>site_repression_min</div> <div><div>ARM</div><div>REAL, SIGNED</div><div>AGG EDGE</div><div>29%</div></div> | Strongest (most negative) predicted repression among the arm's sites (median Å·1.964).                                                                                                                                                                                                                                |
| <div>site_sites_per_gene</div> <div><div>ARM</div><div>REAL Å·Å¥ 0</div><div>29%</div></div>                     | Canonical sites per targeted gene (median 1.34) Å€" site DENSITY, separating an arm with many sites on few genes from one spread thin. <code>site_n_canonical / site_n_genes_canonical</code> .<br>  <i>Defined on: arms in the scanMiR site-type table (771) Å€" Åš HALLMARK-SCOPED, 1,432 genes only</i>            |

**ARB\_ Å· 16**  
supporting Å€" read these when the core raises a question

Per-ARM rollup of its edge set Å€" how many edges and genes it regulates and its mean |beta| over them.

16

 columns Å· coverage **28%Å€"30%** (median 30%) Å· **10** share the block description  
12× count Å· 2× real Å·Å¥ 0 Å· 1× fraction 0Å€"1 Å· 1× signed Å·1Å€"1

|                                                                                                                    |                                                                               |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <div>arb_frac_identified</div> <div><div>ARM</div><div>FRACTION 0Å€"1</div><div>AGG EDGE</div><div>30%</div></div> | <code>arb_n_identified / arb_n_edges</code> (verified exact on 100% of rows). |
| <div>arb_max_abs_beta</div> <div><div>ARM</div><div>REAL Å·Å¥ 0</div><div>AGG EDGE</div><div>30%</div></div>       | Largest  Î²  among the arm's edges.                                           |

|                                                                                                                                                        |                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>arb_max_identity</div> <div><div>ARM</div><div>SIGNED <math>\hat{A}^{\cdot}1\hat{A}\epsilon 1</math></div><div>AGG EDGE</div><div>28%</div></div> | Largest <code>identity</code> among this arm's edges, over reliable rows only. $\hat{A}\epsilon$ ... Now runs $\hat{A}^{\cdot}0.255\hat{A}\epsilon +1.000\hat{A}\epsilon$ ” it reached <b>+740.0</b> until the inert <code>beta_frac_reliable</code> gate was replaced (2026-08-19). |
| <div>arb_mean_abs_beta</div> <div><div>ARM</div><div>REAL <math>\hat{A}\% \neq 0</math></div><div>AGG EDGE</div><div>30%</div></div>                   | Mean $ \hat{I}^2 $ over the arm's edges. See <code>arb_max_abs_beta</code> for the one-edge degeneracy (9.3% of arms, where the two are equal by construction).<br><div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                             |
| <div>arb_n_b001</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                          | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 118</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                        |
| <div>arb_n_b001_ident</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                    | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 51</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_b005</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                          | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 45</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_b005_ident</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                    | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 42</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_b010</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                          | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 28</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_b010_ident</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                    | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 28</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_b025</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                          | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 13</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_b025_ident</div> <div><div>ARM</div><div>COUNT</div><div>30%</div></div>                                                                    | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 13</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_comp_explained</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>30%</div></div>                                             | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 55</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_edges</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>30%</div></div>                                                      | Edges this arm carries in the LEARNED MODEL's own readouts.                                                                                                                                                                                                                          |
| <div>arb_n_identified</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>30%</div></div>                                                 | <div>described by the block above <math>\hat{A}\cdot 0\hat{A}\epsilon</math>” 51</div> <div>  Defined on: arms with a learned-<math>\hat{I}^2</math> edge rollup (728)</div>                                                                                                         |
| <div>arb_n_identity_reliable</div> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>30%</div></div>                                          | Edges of this arm whose <code>identity</code> share is usable ( <code>identity_reliable</code> ).                                                                                                                                                                                    |

HEALTHY\_  $\hat{A}\cdot 3$

supporting  $\hat{A}\epsilon$ ” read these when the core raises a question

Healthy-state legs and flags  $\hat{A}\epsilon$ ” including `healthy_uninformative`, which marks rows where the healthy baseline cannot carry a claim.

3

 columns  $\hat{A}\cdot$  coverage **16% $\hat{A}\epsilon$ ”20%** (median 16%)  $\hat{A}\cdot$  **3** share the block description

1 $\times$  categorical  $\hat{A}\cdot$  1 $\times$  real  $\hat{A}\% \neq 0\hat{A}\cdot$  1 $\times$  boolean

|                                                                                      |                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>healthy_leg</div> <div><div>ARM</div><div>CATEGORICAL</div><div>20%</div></div> | <div>described by the block above <math>\hat{A}\cdot 3</math> levels</div> <div>  Defined on: arms with a GTEx-HEALTHY leg (the thinnest state <math>\hat{A}\epsilon</math>” many arms are multi-mapping-collapsed, MH-166)</div> |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                          |                                                                                                                                                                                  |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>healthy_potential</b><br><div>ARMREAL 016%</div>      | <i>described by the block above 0.27 17.9</i><br><div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>          |
| <b>healthy_uninformative</b><br><div>ARMBOOLEAN16%</div> | <i>described by the block above 7 true / 392 false</i><br><div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div> |

**(BARE) 4**  
supporting read these when the core raises a question

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY\_NAT is the FIELD delta, NAT\_TUM the malignant step.

4 columns coverage 15%100% (median 17%) 3 share the block description  
3x real, signed 1x identifier

|                                                         |                                                                                                                                                                                  |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>arm</b><br><div>KEYIDENTIFIER100%</div>              | KEY mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through <code>_arm_key()</code> , which gates against the canonical universe and LOGS drops rather than silently dropping. |
| <b>dGlobal_HLY_NAT</b><br><div>ARMREAL, SIGNED15%</div> | <i>described by the block above -40 75</i><br><div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>             |
| <b>dGlobal_HLY_TUM</b><br><div>ARMREAL, SIGNED15%</div> | <i>described by the block above -35 75</i><br><div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>             |
| <b>dGlobal_NAT_TUM</b><br><div>ARMREAL, SIGNED20%</div> | <i>described by the block above -49 74</i><br><div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                                    |

**FAME\_ 28**  
supporting read these when the core raises a question

Study-attention axis for the arm distinct PMIDs and distinct curated genes.

28 columns coverage 94%96% (median 96%) 23 share the block description  
27x count 1x fraction 01

|                                                                                 |                                                                                                                                                    |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fame_assay_binding_functional_mti_weak_studies</b><br><div>ARMCOUNT96%</div> | <i>described by the block above 0 928</i><br><div>Defined on: arms with miRTarBase curation / ledger PMIDs FAME axes, not targetome (MH-208)</div> |
| <b>fame_assay_binding_studies</b><br><div>ARMCOUNT96%</div>                     | Studies whose assay class is BINDING (the marginal of the assay functional-class cross-tab).                                                       |
| <b>fame_assay_functional_mti_studies</b><br><div>ARMCOUNT96%</div>              | <i>described by the block above 0 70</i><br><div>Defined on: arms with miRTarBase curation / ledger PMIDs FAME axes, not targetome (MH-208)</div>  |
| <b>fame_assay_functional_mti_weak_studies</b><br><div>ARMCOUNT96%</div>         | <i>described by the block above 0 929</i><br><div>Defined on: arms with miRTarBase curation / ledger PMIDs FAME axes, not targetome (MH-208)</div> |

|                                                                                                               |                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fame_assay_nonfunctional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>           | <i>described by the block above</i> 35<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_other_functional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>        | <i>described by the block above</i> 27<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_other_functional_mti_weak_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>   | <i>described by the block above</i> 247<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208) |
| <b>fame_assay_other_nonfunctional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>     | <i>described by the block above</i> 7<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)   |
| <b>fame_assay_other_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>                       | <i>described by the block above</i> 259<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208) |
| <b>fame_assay_protein_functional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>      | <i>described by the block above</i> 57<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_protein_functional_mti_weak_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div> | <i>described by the block above</i> 25<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_protein_nonfunctional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>   | <i>described by the block above</i> 24<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_protein_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>                     | <i>described by the block above</i> 61<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_reporter_functional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>     | <i>described by the block above</i> 65<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_reporter_nonfunctional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>  | <i>described by the block above</i> 31<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_reporter_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>                    | <i>described by the block above</i> 73<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_rna_functional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>          | <i>described by the block above</i> 57<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |
| <b>fame_assay_rna_functional_mti_weak_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>     | <i>described by the block above</i> 629<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208) |
| <b>fame_assay_rna_nonfunctional_mti_studies</b> <div><div>ARM</div><div>COUNT</div><div>96%</div></div>       | <i>described by the block above</i> 17<br><i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> FAME axes, not targetome (MH-208)  |

|                                                                       |                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fame_assay_rna_studies</b><br><div>ARMCOUNT96%</div>               | <i>described by the block above</i> 0 638<br>  <i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> 638 <i>FAME axes, not targetome (MH-208)</i>                                              |
| <b>fame_assay_total_studies</b><br><div>ARMCOUNT96%</div>             | Distinct STUDIES ( <code>study_id.nunique()</code> ) behind this arm's curated edges median 57, max 930. See <code>fame_assay_unique_targets</code> : different definitions, one axis (spearman 1.0000). |
| <b>fame_assay_unique_targets</b><br><div>ARMCOUNT96%</div>            | Distinct TARGET GENES this arm has in the curated literature ( <code>gene.nunique()</code> ).                                                                                                            |
| <b>fame_led_frac_ltfunc</b><br><div>ARMFRACTION0.011AGG EDGE96%</div> | <i>described by the block above</i> 0 1<br>  <i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> 1 <i>FAME axes, not targetome (MH-208)</i>                                                  |
| <b>fame_led_n_classes</b><br><div>ARMCOUNTAGG EDGE96%</div>           | How many distinct evidence CLASSES this arm's curated edges span (1-7).                                                                                                                                  |
| <b>fame_led_n_ltfunc_pmid</b><br><div>ARMCOUNTAGG EDGE96%</div>       | <i>described by the block above</i> 0 197<br>  <i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> 197 <i>FAME axes, not targetome (MH-208)</i>                                              |
| <b>fame_led_n_weak</b><br><div>ARMCOUNTAGG EDGE94%</div>              | <i>described by the block above</i> 1 754<br>  <i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> 754 <i>FAME axes, not targetome (MH-208)</i>                                              |
| <b>fame_n_genes_curated</b><br><div>ARMCOUNTAGG EDGE96%</div>         | <i>described by the block above</i> 1 3615<br>  <i>Defined on: arms with miRTarBase curation / ledger PMIDs</i> 3615 <i>FAME axes, not targetome (MH-208)</i>                                            |
| <b>fame_npmid</b><br><div>ARMCOUNTAGG EDGE96%</div>                   | Distinct PMIDs citing this arm.                                                                                                                                                                          |

AGO\_ 4

supporting read these when the core raises a question

4 columns coverage 28-60% (median 28%)  
1x fraction 0.01 1x constant 1x categorical 1x real 0

|                                                        |                                                                                                                                                                      |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ago_dom</b><br><div>ARMFRACTION0.01128%</div>       | Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. |
| <b>ago_dom_src</b><br><div>ARMCONSTANT28%</div>        | Provenance of <code>ago_dom</code> which CLIP dataset the AGO-loading dominance came from.                                                                           |
| <b>ago_guide_class</b><br><div>ARMCATEGORICAL28%</div> | AGO/RISC loading gate for the arm.                                                                                                                                   |
| <b>ago_reads</b><br><div>ARMREAL060%</div>             | Raw AGO-CLIP read count (median 6, max 210,722).                                                                                                                     |

HLY\_ 7

supporting read these when the core raises a question

HEALTHY baseline levels for the arm GTEx breast median/percentile and TCGA-NAT median.  
7 columns coverage 17-91% (median 91%) 5 share the block description  
2x real 0 1x categorical 1x boolean 1x percent 1x count 1x fraction 0.01

|                                                                                           |                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>hly_baseline_src</div> <div><div>ARM</div><div>CATEGORICAL</div><div>91%</div></div> | <div>described by the block above</div> <div>4 levels</div> <div>Defined on: arms with a GTEx healthy anchor "â€"â€š MEASURED leg only; floor0 is NEVER emitted as a value</div>              |
| <div>hly_from_seedmate</div> <div><div>ARM</div><div>BOOLEAN</div><div>91%</div></div>    | <div>described by the block above</div> <div>49 true / 2,188 false</div> <div>Defined on: arms with a GTEx healthy anchor "â€"â€š MEASURED leg only; floor0 is NEVER emitted as a value</div> |
| <div>hly_gtex_median</div> <div><div>ARM</div><div>REAL</div><div>17%</div></div>         | <div>described by the block above</div> <div>0.0803 " 18.1</div> <div>Defined on: arms with a GTEx healthy anchor "â€"â€š MEASURED leg only; floor0 is NEVER emitted as a value</div>         |
| <div>hly_gtex_pct</div> <div><div>ARM</div><div>PERCENT</div><div>17%</div></div>         | <div>described by the block above</div> <div>0.0514 " 1</div> <div>Defined on: arms with a GTEx healthy anchor "â€"â€š MEASURED leg only; floor0 is NEVER emitted as a value</div>            |
| <div>hly_nat_detect_n</div> <div><div>ARM</div><div>COUNT</div><div>91%</div></div>       | <div>In how many of the ~104 matched NAT samples this arm is detected above the floor.</div>                                                                                                  |
| <div>hly_nat_median</div> <div><div>ARM</div><div>REAL</div><div>74%</div></div>          | <div>Median NAT abundance (fill 74.0%).</div>                                                                                                                                                 |
| <div>hly_tumor_prev</div> <div><div>ARM</div><div>FRACTION</div><div>91%</div></div>      | <div>described by the block above</div> <div>0 " 1</div> <div>Defined on: arms with a GTEx healthy anchor "â€"â€š MEASURED leg only; floor0 is NEVER emitted as a value</div>                 |

DOSE\_ 3

supporting " read these when the core raises a question

|                                                                                                                                                                                                                                                                                |                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.</div> <div>3 columns " coverage 12%â€"12% (median 12%) " 3 share the block description</div> <div>2x real " " 1x constant</div> |                                                                                                                                                |
| <div>dose_comp_retention</div> <div><div>ARM</div><div>REAL</div><div>12%</div></div>                                                                                                                                                                                          | <div>described by the block above</div> <div>0.51 " 1.66</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div> |
| <div>dose_confounded</div> <div><div>ARM</div><div>CONSTANT</div><div>12%</div></div>                                                                                                                                                                                          | <div>described by the block above</div> <div>= False</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>     |
| <div>dose_prolif_retention</div> <div><div>ARM</div><div>REAL</div><div>12%</div></div>                                                                                                                                                                                        | <div>described by the block above</div> <div>0.69 " 1.25</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div> |

ABUND\_ 5

supporting " read these when the core raises a question

|                                                                                                                                                                                                                                                      |                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <div>The arm's own abundance profile across the cohort " median, SD, IQR, detection fraction.</div> <div>5 columns " coverage 88%â€"91% (median 91%) " 4 share the block description</div> <div>3x real " " 1x fraction "â€"1 " 1x categorical</div> |                                                                                                                                         |
| <div>abund_frac_detected</div> <div><div>ARM</div><div>FRACTION</div><div>91%</div></div>                                                                                                                                                            | <div>described by the block above</div> <div>0 " 1</div> <div>Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)</div> |

|                                                                     |                                                                                                                                       |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>abund_iqr</b>                                                    | described by the block above $\hat{A} \cdot 0 \hat{A} \epsilon'' 4.59$                                                                |
| <b>ARM</b> <b>REAL <math>\hat{A} \epsilon'' 0</math></b> <b>91%</b> | Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)                                                                   |
| <b>abund_med</b>                                                    | described by the block above $\hat{A} \cdot 0.105 \hat{A} \epsilon'' 17.9$                                                            |
| <b>ARM</b> <b>REAL <math>\hat{A} \epsilon'' 0</math></b> <b>91%</b> | Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)                                                                   |
| <b>abund_sd</b>                                                     | described by the block above $\hat{A} \cdot 0.00545 \hat{A} \epsilon'' 3.11$                                                          |
| <b>ARM</b> <b>REAL <math>\hat{A} \epsilon'' 0</math></b> <b>88%</b> | Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)                                                                   |
| <b>abund_sd_tertile</b>                                             | Dispersion tertile $\hat{A} \epsilon'' 718 / 716 / 718$ by construction, so it carries no information beyond <b>abund_sd</b> 's rank. |
| <b>ARM</b> <b>CATEGORICAL</b> <b>88%</b>                            |                                                                                                                                       |

**ATTR\_  $\hat{A} \cdot 14$**   
reference  $\hat{A} \epsilon''$  context, provenance and rarely-quoted detail

Attribution rollup over the arm's edges  $\hat{A} \epsilon''$  how much of the gene's coupling this arm is credited with, and over how many reliable edges.

**14** columns  $\hat{A} \cdot$  coverage **6% $\hat{A} \epsilon'' 29\%$**  (median 15%)  $\hat{A} \cdot$  **9** share the block description

11x fraction  $0 \hat{A} \epsilon'' 1 \hat{A} \cdot$  2x count  $\hat{A} \cdot$  1x signed  $\hat{A} \cdot 1 \hat{A} \epsilon'' 1$

|                                                                                       |                                                                                                                                    |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>attr_max_budget_share</b>                                                          | described by the block above $\hat{A} \cdot 7.64e-06 \hat{A} \epsilon'' 0.997$                                                     |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_max_credit_share</b>                                                          | described by the block above $\hat{A} \cdot 0.000114 \hat{A} \epsilon'' 0.996$                                                     |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>6%</b>              | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_max_joint_share</b>                                                           | Largest JOINT attribution share among this arm's edges.                                                                            |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             |                                                                                                                                    |
| <b>attr_max_nested_share</b>                                                          | Largest NESTED attribution share. See <b>attr_max_joint_share</b> : different values (agree on 35.4%), one axis (spearman +0.962). |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_max_within_share</b>                                                          | Largest WITHIN-cell attribution share for this arm.                                                                                |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             |                                                                                                                                    |
| <b>attr_med_budget_share</b>                                                          | described by the block above $\hat{A} \cdot 7.64e-06 \hat{A} \epsilon'' 0.521$                                                     |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_med_credit_share</b>                                                          | described by the block above $\hat{A} \cdot 0.000114 \hat{A} \epsilon'' 0.996$                                                     |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>6%</b>              | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_med_dose_share</b>                                                            | Median dose share of this arm within its family cells.                                                                             |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>29%</b>             |                                                                                                                                    |
| <b>attr_med_joint_share</b>                                                           | described by the block above $\hat{A} \cdot 0 \hat{A} \epsilon'' 0.645$                                                            |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_med_nested_share</b>                                                          | described by the block above $\hat{A} \cdot 0 \hat{A} \epsilon'' 0.692$                                                            |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_med_phi_joint</b>                                                             | described by the block above $\hat{A} \cdot -0.0035 \hat{A} \epsilon'' +0.119$                                                     |
| <b>ARM</b> <b>SIGNED <math>\hat{A} \cdot 1 \hat{A} \epsilon'' 1</math></b> <b>15%</b> | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |
| <b>attr_med_within_share</b>                                                          | described by the block above $\hat{A} \cdot 0 \hat{A} \epsilon'' 1$                                                                |
| <b>ARM</b> <b>FRACTION <math>0 \hat{A} \epsilon'' 1</math></b> <b>15%</b>             | Defined on: arms with a Shapley/attribution decomposition over their edges (728)                                                   |

|                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>attr_n_edges</div> <div><div>ARM</div><div>COUNT</div><div>22%</div></div>                                                                                                                                                                                                                                                       | <div>described by the block above</div> <div>1 50</div> <div>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</div>                                  |
| <div>attr_n_reliable</div> <div><div>ARM</div><div>COUNT</div><div>15%</div></div>                                                                                                                                                                                                                                                    | <div>Reliable edges behind this arm's</div> <div>attr_*</div> <div>shares (median 2).</div>                                                                                          |
| <div>CTX_ 2</div> <div>reference context, provenance and rarely-quoted detail</div>                                                                                                                                                                                                                                                   |                                                                                                                                                                                      |
| <div>Gene-design context from the decoy/atlas layer ( learned/analyses/gene_atlas.py via card_context ): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.</div> <div>2 columns coverage 29%29% (median 29%) 2 share the block description</div> <div>1x boolean 1x real 0</div> |                                                                                                                                                                                      |
| <div>ctx_arm_abundant</div> <div><div>ARM</div><div>BOOLEAN</div><div>29%</div></div>                                                                                                                                                                                                                                                 | <div>described by the block above</div> <div>213 true / 499 false</div> <div>Defined on: arms present on the EDGE card (729 of 2,450) lifted arm-rung columns, names preserved</div> |
| <div>ctx_arm_dose</div> <div><div>ARM</div><div>REAL 0</div><div>29%</div></div>                                                                                                                                                                                                                                                      | <div>described by the block above</div> <div>0.133 17.9</div> <div>Defined on: arms present on the EDGE card (729 of 2,450) lifted arm-rung columns, names preserved</div>           |
| <div>GRANK_ 3</div> <div>reference context, provenance and rarely-quoted detail</div>                                                                                                                                                                                                                                                 |                                                                                                                                                                                      |
| <div>The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.</div> <div>3 columns coverage 15%20% (median 20%) 3 share the block description</div> <div>3x count</div>                                                                                      |                                                                                                                                                                                      |
| <div>grank_HLY</div> <div><div>ARM</div><div>COUNT</div><div>15%</div></div>                                                                                                                                                                                                                                                          | <div>described by the block above</div> <div>24 100</div> <div>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</div>     |
| <div>grank_NAT</div> <div><div>ARM</div><div>COUNT</div><div>20%</div></div>                                                                                                                                                                                                                                                          | <div>described by the block above</div> <div>0 100</div> <div>Defined on: edges with a matched NAT leg (n~104 paired participants)</div>                                             |
| <div>grank_TUM</div> <div><div>ARM</div><div>COUNT</div><div>20%</div></div>                                                                                                                                                                                                                                                          | <div>described by the block above</div> <div>3 100</div> <div>Defined on: edges present in the progression panel (arm expressed + state measured)</div>                              |
| <div>REAL_ 11</div> <div>reference context, provenance and rarely-quoted detail</div>                                                                                                                                                                                                                                                 |                                                                                                                                                                                      |
| <div>Realization rollup for the ARM how many of its edges are scored, and its median and best realized coupling.</div> <div>11 columns coverage 9%24% (median 24%) 8 share the block description</div> <div>7x count 2x signed 1 2x fraction 01</div>                                                                                 |                                                                                                                                                                                      |
| <div>real_best_coupling</div> <div><div>ARM</div><div>SIGNED 11</div><div>24%</div></div>                                                                                                                                                                                                                                             | <div>Most negative realized coupling among this arm's scored genes (median 0.08, min 0.56).</div>                                                                                    |
| <div>real_frac_c10</div> <div><div>ARM</div><div>FRACTION 01</div><div>9%</div></div>                                                                                                                                                                                                                                                 | <div>Fraction of this arm's scored genes coupling below 0.10.</div>                                                                                                                  |

|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>real_frac_sig</div> <div><div>ARM</div><div>FRACTION 0â€"1</div><div>9%</div></div>                                  | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 1</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                                |
| <div>real_med_coupling</div> <div><div>ARM</div><div>SIGNED <math>\hat{A}^{'1}\hat{a}€ 1</math></div><div>24%</div></div> | <div>described by the block above <math>\hat{A} \cdot -0.418 \hat{a}€" +0.222</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                      |
| <div>real_n_c05</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 43</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                               |
| <div>real_n_c10</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 37</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                               |
| <div>real_n_c15</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 32</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                               |
| <div>real_n_c20</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 30</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                               |
| <div>real_n_c30</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 25</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                               |
| <div>real_n_scored</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                          | <div>Genes scored for this arm's realized coupling (median 3). <math>\hat{a}œ</math>... Its <code>n_c05</code> <math>\hat{a}€ </math> <code>n_c30</code> ladder is properly nested <math>\hat{a}€"</math> <b>0 violations</b> (verified).</div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div> |
| <div>real_n_sig</div> <div><div>ARM</div><div>COUNT</div><div>24%</div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{a}€" 32</math></div> <div>Defined on: arms with <math>\hat{a}\%¥1</math> calibrated-coupling-scored edge (593) <math>\hat{a}€"</math> the realization LADDER; rates are gated at <math>n\hat{a}\%¥5</math></div>                                                                                                                                                               |

TS\_  $\hat{A} \cdot 10$

reference  $\hat{a}€"$  context, provenance and rarely-quoted detail

TargetScan context++ scores for the arm's sites  $\hat{a}€"$  number of sites, mean and best context score. More negative context++ = stronger predicted repression.

10 columns  $\hat{A} \cdot$  coverage **13%** $\hat{a}€"$ **13%** (median 13%)  $\hat{A} \cdot$  **8** share the block description

5× count  $\hat{A} \cdot$  2× real, signed  $\hat{A} \cdot$  1× signed  $\hat{a}^{'1}\hat{a}€|1$   $\hat{A} \cdot$  1× fraction 0 $\hat{a}€"$ 1  $\hat{A} \cdot$  1× real  $\hat{a}\%¥0$

|                                                                                                                     |                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>ts_ctx_best</div> <div><div>ARM</div><div>REAL, SIGNED</div><div>13%</div></div>                               | <div>Best (most negative) TargetScan context++ score for this arm <math>\hat{a}€"</math> range [<math>\hat{a}^{'2.02}</math>, <math>\hat{a}^{'0.34}</math>], <b>all negative by construction</b> (context++ scores repression as a negative number).</div> |
| <div>ts_ctx_mean</div> <div><div>ARM</div><div>SIGNED <math>\hat{A}^{'1}\hat{a}€ 1</math></div><div>13%</div></div> | <div>described by the block above <math>\hat{A} \cdot -0.445 \hat{a}€" -0.0356</math></div> <div>Defined on: arms in TargetScan's human default predictions (<math>\hat{a}\\$</math> only 321)</div>                                                       |
| <div>ts_frac_8mer</div> <div><div>ARM</div><div>FRACTION 0â€"1</div><div>13%</div></div>                            | <div>described by the block above <math>\hat{A} \cdot 0.0335 \hat{a}€" 0.64</math></div> <div>Defined on: arms in TargetScan's human default predictions (<math>\hat{a}\\$</math> only 321)</div>                                                          |
| <div>ts_kd_med</div> <div><div>ARM</div><div>REAL, SIGNED</div><div>13%</div></div>                                 | <div>described by the block above <math>\hat{A} \cdot -5.64 \hat{a}€" -2.03</math></div> <div>Defined on: arms in TargetScan's human default predictions (<math>\hat{a}\\$</math> only 321)</div>                                                          |

|                                                                                              |                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ts_n_7mer_A1</b> <div><div>ARM</div><div>COUNT</div><div>13%</div></div>                  | <i>described by the block above</i> 0 1100<br><i>Defined on: arms in TargetScan's human default predictions (â€” only 321)</i>                                                                |
| <b>ts_n_7mer_m8</b> <div><div>ARM</div><div>COUNT</div><div>13%</div></div>                  | <i>described by the block above</i> 0 1352<br><i>Defined on: arms in TargetScan's human default predictions (â€” only 321)</i>                                                                |
| <b>ts_n_8mer</b> <div><div>ARM</div><div>COUNT</div><div>13%</div></div>                     | <i>described by the block above</i> 2 1265<br><i>Defined on: arms in TargetScan's human default predictions (â€” only 321)</i>                                                                |
| <b>ts_n_genes</b> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>13%</div></div> | Genes with a TargetScan context++ site  median 497, fill <b>12.9%, the SPARSEST universe on the card</b> . Correlates +0.726 with the MANE seed scan and only <b>+0.231</b> with scanMiR-any. |
| <b>ts_n_sites</b> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>13%</div></div> | <i>described by the block above</i> 5 2475<br><i>Defined on: arms in TargetScan's human default predictions (â€” only 321)</i>                                                                |
| <b>ts_sites_per_gene</b> <div><div>ARM</div><div>REAL 0</div><div>13%</div></div>            | <i>described by the block above</i> 1 1.27<br><i>Defined on: arms in TargetScan's human default predictions (â€” only 321)</i>                                                                |

**ISOC\_ 5**

reference context, provenance and rarely-quoted detail

isomiR COMPOSITION what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm.

5 columns coverage **91%91%** (median 91%) 4 share the block description  
3x fraction 01 1x real 0 1x count

|                                                                                          |                                                                                                                                                                            |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>isoc_canonical_frac</b> <div><div>ARM</div><div>FRACTION 01</div><div>91%</div></div> | <i>described by the block above</i> 0.151 1<br><i>Defined on: arms in the isomiR seed-COMPOSITION table  where the reads go; â isoc_orphan_frac flags MH-153 mass loss</i> |
| <b>isoc_donated_frac</b> <div><div>ARM</div><div>FRACTION 01</div><div>91%</div></div>   | <i>described by the block above</i> 0 0.5<br><i>Defined on: arms in the isomiR seed-COMPOSITION table  where the reads go; â isoc_orphan_frac flags MH-153 mass loss</i>   |
| <b>isoc_med_reads</b> <div><div>ARM</div><div>REAL 0</div><div>91%</div></div>           | Median isomiR reads for this arm <b>median 1.00, max 528,927</b> .                                                                                                         |
| <b>isoc_n_dest_families</b> <div><div>ARM</div><div>COUNT</div><div>91%</div></div>      | <i>described by the block above</i> 0 6<br><i>Defined on: arms in the isomiR seed-COMPOSITION table  where the reads go; â isoc_orphan_frac flags MH-153 mass loss</i>     |
| <b>isoc_orphan_frac</b> <div><div>ARM</div><div>FRACTION 01</div><div>91%</div></div>    | <i>described by the block above</i> 0 0.823<br><i>Defined on: arms in the isomiR seed-COMPOSITION table  where the reads go; â isoc_orphan_frac flags MH-153 mass loss</i> |

**SDSZ\_ 4**

reference context, provenance and rarely-quoted detail

MANE 3'UTR seed scan.

4 columns coverage **97%97%** (median 97%) 3 share the block description  
2x count 1x real 0 1x fraction 01

|                                                                                                    |                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>sdsz_N_7mer_plus</b> <div><div>ARM</div><div>COUNT</div><div>AGG EDGE</div><div>97%</div></div> | <i>described by the block above</i> 102 8814<br><i>Defined on: arms in the MANE 3'UTR seed scan  â€” â€” a FOURTH targetome universe, never blend (2,370)</i> |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

ARM COUNT AGG EDGE 97%

described by the block above  $\hat{A} \cdot 8 \hat{\epsilon}''$  3030

*Defined on: arms in the MANE 3'UTR seed scan â€” a FOURTH targetome universe, never blend (2.370)*

sdsz\_N\_eff

ARM REAL 0 AGG EDGE

97%

Effective seed-site count over the scanned universe (median 2,094, fill **96.7%** the best-covered targetome axis).

sdsz\_rarity

ARM FRACTION 0.1 97%

described by the block above  $\hat{A} \cdot 0 \in 1$

*Defined on: arms in the MANE 3'UTR seed scan â€” a FOURTH targetome universe, never blend (2,370)*

CHIM\_ 5

reference " context, provenance and rarely-quoted detail

Chimeric-evidence ROLLUP for the arm (aggregated over its edges).

```
5 columns Â· coverage 21%â€“63% (median 60%) Â· 5 share the block description
4x count Â· 1x real â†¥ 0
```

chim\_manakov\_n\_genes

ARM COUNT AGG EDGE 60%

described by the block above  $\hat{A} \cdot 0 \in \mathbb{R}^{13825}$

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim manakov\_weight

ARM REAL  $\hat{\Delta} \approx 0$  AGG EDGE

60%

described by the block above  $\hat{A} \cdot 1 \hat{\epsilon}'' 2.11\text{e}+05$

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim\_n\_sources

ARM COUNT AGG EDGE 63%

described by the block above  $\hat{A} \cdot 1 \in \mathbb{N}$

Defined on: arms with chimeric ligation evidence (1,538)  $\hat{a} \hat{e}^* \hat{a} \hat{s} \sim 90\%$  Manakov/HEK293T, per-source, NEVER pooled

chim\_tarbase\_n\_genes

ARM COUNT AGG EDGE 21%

described by the block above  $\hat{A} \cdot 1 \in \mathbb{Z}^{2701}$

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim\_tarbase\_weight

|     |       |     |      |     |
|-----|-------|-----|------|-----|
| ARM | COUNT | AGG | EDGE | 21% |
|-----|-------|-----|------|-----|

described by the block above  $\hat{A} \cdot 1 \hat{\epsilon}''$  10255

*Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled*

FST\_ 16

reference " context, provenance and rarely-quoted detail

â THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES â€” gene-free (`card_ladders.family_state_shift`, 2026-08-19).

16 columns Â· coverage 3%â€”100% (median 20%) Â· 8 share the block description  
6x count Â· 5x boolean Â· 3x fraction 0â€”1 Â· 2x signed â€”1â€”1

fst\_d\_share\_HLY\_TUM

ARM SIGNED  $\hat{A}^{-1} \in \mathbb{R}^{1 \times 1}$  3%

As `fst_d_share_NAT_TUM` but GTEEx-healthy  $\hat{\pi}^*$  tumour, same common-support correction.  $n=68$  (HLY is the sparsest leg at 17.0%). Post-fix **mean  $\hat{\pi}^*$  0.0000, 47.1% positive,  $p=0.93$ .**

fst\_d\_share\_NAT\_TUM

ARM SIGNED 6%

Change in the arm's share of its seed family's linear dose, matched NAT  $\hat{\alpha}_i^*$  tumour, **re-normalised over the members measured in BOTH states.**

|                                |                                                                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fst_dominance_switch</b>    | <i>described by the block above</i> $\hat{A} \cdot 103$ true / 74 false                                                                                                                                                           |
| <b>fst_hly_family_measured</b> | Do ALL members of this arm's family have a measured GTEx level.                                                                                                                                                                   |
| <b>fst_hly_measured</b>        | <i>described by the block above</i> $\hat{A} \cdot 417$ true / 2,033 false                                                                                                                                                        |
| <b>fst_is_dominant_HLY</b>     | <i>described by the block above</i> $\hat{A} \cdot 345$ true / 72 false                                                                                                                                                           |
| <b>fst_is_dominant_TUM</b>     | <i>described by the block above</i> $\hat{A} \cdot 394$ true / 91 false                                                                                                                                                           |
| <b>fst_n_common_HLY_TUM</b>    | Members measured in BOTH GTEx-healthy and tumour $\hat{A}$ the denominator behind <code>fst_d_share_HLY_TUM</code> . See <code>fst_n_common_NAT_TUM</code> .                                                                      |
| <b>fst_n_common_NAT_TUM</b>    | How many of the family's arms are measured in BOTH NAT and tumour $\hat{A}$ the DENOMINATOR behind <code>fst_d_share_NAT_TUM</code> , shipped because the column was wrong for want of it.                                        |
| <b>fst_n_members</b>           | <i>described by the block above</i> $\hat{A} \cdot 1$ $\hat{A}$ 22                                                                                                                                                                |
| <b>fst_rank_HLY</b>            | <i>described by the block above</i> $\hat{A} \cdot 1$ $\hat{A}$ 6                                                                                                                                                                 |
| <b>fst_rank_NAT</b>            | <i>described by the block above</i> $\hat{A} \cdot 1$ $\hat{A}$ 10                                                                                                                                                                |
| <b>fst_rank_TUM</b>            | <i>described by the block above</i> $\hat{A} \cdot 1$ $\hat{A}$ 8                                                                                                                                                                 |
| <b>fst_share_HLY</b>           | Share of family dose in GTEx-healthy, from <code>hly_gtex_median</code> (fill <b>17.0%</b> $\hat{A}$ the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only.                                          |
| <b>fst_share_NAT</b>           | Share of family dose in matched NAT, from <code>hly_nat_median</code> (fill <b>74.0%</b> $\hat{A}$ the best-covered leg, 3.7 $\hat{A}$ TUM). Same singleton degeneracy: median 1.000, mask on <code>fst_n_members &gt; 1</code> . |
| <b>fst_share_TUM</b>           | The arm's fraction of its seed family's total linear dose in tumour, from <code>arm_med_rpm</code> (fill <b>19.8%</b> ).                                                                                                          |

SUB\_  $\hat{A} \cdot 6$

reference  $\hat{A}$  context, provenance and rarely-quoted detail

PAM50 subtype block aggregated to the ARM. See `esub_` for the per-edge form.

**6** columns  $\hat{A} \cdot$  coverage **30%** $\hat{A}$ **30%** (median 30%)  $\hat{A} \cdot$  **4** share the block description  
2 $\times$  count  $\hat{A} \cdot$  2 $\times$  signed  $\hat{A}^{\wedge}1$  $\hat{A}$  1 $\times$  categorical  $\hat{A} \cdot$  1 $\times$  fraction  $0\hat{A}$  $\hat{A}$ 1

|                      |                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>sub_best_frac</b> | <i>described by the block above</i> $\hat{A} \cdot 0.25$ $\hat{A}$ 1                                                                                           |
| <b>sub_best_frac</b> | <i>Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) <math>\hat{A}</math> <math>\hat{A}</math> DISTRIBUTIONAL (MH-165), not a validated label</i> |

|                     |                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| sub_best_mode       | described by the block above                                                                                     |
| ARM CATEGORICAL 30% | Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) as DISTRIBUTIONAL (MH-165), not a validated label |
| sub_med_rho_best    | Median coupling in the arm's BEST subtype.                                                                       |
| ARM SIGNED 30%      |                                                                                                                  |
| sub_med_rho_pool    | Median pooled-subtype coupling (median 0.0064) the unselected reference for sub_med_rho_best .                   |
| ARM SIGNED 30%      | Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) as DISTRIBUTIONAL (MH-165), not a validated label |
| sub_n_edges         | described by the block above                                                                                     |
| ARM COUNT 30%       | Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) as DISTRIBUTIONAL (MH-165), not a validated label |
| sub_n_q05           | described by the block above                                                                                     |
| ARM COUNT 30%       | Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) as DISTRIBUTIONAL (MH-165), not a validated label |

SEQ\_ 4  
reference context, provenance and rarely-quoted detail

|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| SEQUENCE targetome degree genes with a strong predicted duplex (scanMiR RBNS K_D), breast-expressed. |
| 4 columns coverage 30%30% (median 30%) 2 share the block description                                 |
| 2x count 2x real 0                                                                                   |

|                        |                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| seq_n_genes_any        | Genes with ANY scanMiR K_D site                                                                                                                               |
| ARM COUNT AGG EDGE 30% |                                                                                                                                                               |
| seq_n_genes_strong     | Genes with a STRONG scanMiR K_D site median 3,632, spearman +0.822 with sdsz_N_eff and +0.797 with site_n_genes_canonical , but only +0.566 with ts_n_genes . |
| ARM COUNT AGG EDGE 30% |                                                                                                                                                               |
| seq_promisc            | described by the block above                                                                                                                                  |
| ARM REAL 30%           | Defined on: arms in the scanMiR GENOME-WIDE K_D scan (746; breast-expressed genes)                                                                            |
| seq_promisc_any        | described by the block above                                                                                                                                  |
| ARM REAL 30%           | Defined on: arms in the scanMiR GENOME-WIDE K_D scan (746; breast-expressed genes)                                                                            |

CUR\_ 2  
reference context, provenance and rarely-quoted detail

|                                        |
|----------------------------------------|
| 2 columns coverage 36%36% (median 36%) |
| 2x count                               |

|                        |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| cur_he_degree          | How many HE genes this arm regulates in the curated design the arm's curation degree (median 3, max 125). |
| ARM COUNT AGG EDGE 36% | Defined on: arms with miRTarBase curation / ledger PMIDs as FAME axes, not targetome (MH-208)             |
| cur_he_degree_expr     | As cur_he_degree but counting only targets above the expression floor.                                    |
| ARM COUNT AGG EDGE 36% |                                                                                                           |

COV\_ 22  
reference “ context, provenance and rarely-quoted detail

Coverage flags “ was this row ever SCANNED by the named source.  
22 columns coverage 100% “100% (median 100%) 21 share the block description  
22× boolean

|             |                                                                                                         |      |
|-------------|---------------------------------------------------------------------------------------------------------|------|
| cov_ago_dom | described by the block above Â·678 true / 1,772 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_attr    | described by the block above Â·546 true / 1,904 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_beta    | described by the block above Â·728 true / 1,722 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_cnv     | described by the block above Â·856 true / 1,594 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_comp    | described by the block above Â·205 true / 2,245 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_cptac   | described by the block above Â·2,071 true / 379 false<br>  Defined on: arms measurable in CPTAC (2,095) |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_curated | described by the block above Â·875 true / 1,575 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_expr    | described by the block above Â·2,231 true / 219 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_famrole | described by the block above Â·2,231 true / 219 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_field   | described by the block above Â·571 true / 1,879 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_focal   | described by the block above Â·733 true / 1,717 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_fst     | Was the family-share block computed for this arm at all “ i.e. did it have the abundance input.         |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_gctx    | described by the block above Â·714 true / 1,736 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_isocomp | described by the block above Â·2,222 true / 228 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_isomir  | described by the block above Â·687 true / 1,763 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_meth    | described by the block above Â·343 true / 2,107 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |
| cov_real    | described by the block above Â·590 true / 1,860 false                                                   |      |
| ARM         | BOOLEAN                                                                                                 | 100% |

|                |                                                       |      |
|----------------|-------------------------------------------------------|------|
| cov_seed_scan  | described by the block above Â·2,370 true / 80 false  |      |
| ARM            | BOOLEAN                                               | 100% |
| cov_seq_scan   | described by the block above Â·736 true / 1,714 false |      |
| ARM            | BOOLEAN                                               | 100% |
| cov_site_types | described by the block above Â·718 true / 1,732 false |      |
| ARM            | BOOLEAN                                               | 100% |
| cov_subtype    | described by the block above Â·728 true / 1,722 false |      |
| ARM            | BOOLEAN                                               | 100% |
| cov_targetscan | described by the block above Â·317 true / 2,133 false |      |
| ARM            | BOOLEAN                                               | 100% |

ISO\_ Â· 5

reference â€” context, provenance and rarely-quoted detail

isomiR profile for the arm â€” 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X\_fam, which is BUILT but default-OFF (coupling wash, bidirectional).

5 columns Â· coverage 28%â€”28% (median 28%) Â· 4 share the block description  
2x fraction 0â€”1 Â· 1x count Â· 1x real â‰¥ 0 Â· 1x real, signed

|                     |                                                                                                                                                                                                                             |     |                                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| iso_n_samples       | described by the block above $\hat{A} \cdot 3 \hat{\in}'' 1078$                                                                                                                                                             |     |                                                                                                                                             |
| ARM                 | COUNT                                                                                                                                                                                                                       | 28% | Defined on: arms with a 5'-isomiR seed-shift measurement (687) $\hat{\in}''$ $\hat{\in}''$ gate on iso_n_samples, the median is n-DEPENDENT |
| iso_rpm_summed      | $\hat{\in}''$ A SUM ACROSS SAMPLES, not a per-sample RPM $\hat{\in}''$ the name misleads. Max 270,988,199, which is impossible for a per-million unit; divided by iso_n_samples the scale is sane (median 57, max 251,381). |     |                                                                                                                                             |
| ARM                 | REAL $\hat{\in}''$ 0                                                                                                                                                                                                        | 28% |                                                                                                                                             |
| iso_seed_shift_frac | described by the block above $\hat{A} \cdot 0 \hat{\in}'' 0.868$                                                                                                                                                            |     |                                                                                                                                             |
| ARM                 | FRACTION $0\hat{\in}''1$                                                                                                                                                                                                    | 28% | Defined on: arms with a 5'-isomiR seed-shift measurement (687) $\hat{\in}''$ $\hat{\in}''$ gate on iso_n_samples, the median is n-DEPENDENT |
| iso_seed_shift_sd   | described by the block above $\hat{A} \cdot 0 \hat{\in}'' 0.354$                                                                                                                                                            |     |                                                                                                                                             |
| ARM                 | FRACTION $0\hat{\in}''1$                                                                                                                                                                                                    | 28% | Defined on: arms with a 5'-isomiR seed-shift measurement (687) $\hat{\in}''$ $\hat{\in}''$ gate on iso_n_samples, the median is n-DEPENDENT |
| iso_top_shift_nt    | described by the block above $\hat{A} \cdot -6 \hat{\in}'' 6$                                                                                                                                                               |     |                                                                                                                                             |
| ARM                 | REAL, SIGNED                                                                                                                                                                                                                | 28% | Defined on: arms with a 5'-isomiR seed-shift measurement (687) $\hat{\in}''$ $\hat{\in}''$ gate on iso_n_samples, the median is n-DEPENDENT |

FIELD\_ Â· 5

reference â€” context, provenance and rarely-quoted detail

Within-patient NAT field-effect fit â€” is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset.

5 columns Â· coverage 23%â€”23% (median 23%) Â· 3 share the block description  
4x signed âˆ’1â€”1 Â· 1x count

|                        |                                                                                                                                                                                                                      |     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| field_excess_over_perm | âˆ’âˆ’ NOT A RETENTION RATIO â€” a FIFTH sense of the word in this codebase. Measured:<br>it equals <code>field_r_own - field_r_perm</code> (exact within rounding on 100% of 571 arms, max<br>residual 0.001), i.e. |     |
| ARM                    | SIGNED $\hat{A}^{-1} \hat{A} \epsilon   1$                                                                                                                                                                           | 23% |

|                                                                                             |                                                                                                                                                          |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>field_n</b> <div><div>ARM</div><div>COUNT</div><div>23%</div></div>                      | <i>described by the block above</i> <code>40</code> <code>103</code> <div>Defined on: arms with a within-patient NAT field-effect fit (571)</div>        |
| <b>field_r_adj</b> <div><div>ARM</div><div>SIGNED <code>1</code></div><div>23%</div></div>  | <i>described by the block above</i> <code>-0.344</code> <code>+0.665</code> <div>Defined on: arms with a within-patient NAT field-effect fit (571)</div> |
| <b>field_r_own</b> <div><div>ARM</div><div>SIGNED <code>1</code></div><div>23%</div></div>  | <i>described by the block above</i> <code>-0.304</code> <code>+0.624</code> <div>Defined on: arms with a within-patient NAT field-effect fit (571)</div> |
| <b>field_r_perm</b> <div><div>ARM</div><div>SIGNED <code>1</code></div><div>23%</div></div> | The permutation null for <code>field_r_own</code> median <b>0.000</b> , range [ <code>0.11</code> , <code>+0.09</code> ].                                |

**FAMROLE\_ 6**  
reference context, provenance and rarely-quoted detail

The arm's ROLE inside its seed family member count, its share of family abundance, and its rank among members.

**6** columns `coverage` **88%** (median 91%) **5** share the block description  
3× count `1` `fraction` `1` `1` categorical `1` boolean

|                                                                                                      |                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>famrole_abund_rank</b> <div><div>ARM</div><div>COUNT</div><div>91%</div></div>                    | <i>described by the block above</i> <code>1</code> <code>14</code> <div>Defined on: arms with a seed family the arm's ROLE inside it; <code>DEGENERATE</code> at <code>famrole_n_members==1</code></div>                                             |
| <b>famrole_abund_share</b> <div><div>ARM</div><div>FRACTION <code>1</code></div><div>91%</div></div> | <i>described by the block above</i> <code>0</code> <code>1</code> <div>Defined on: arms with a seed family the arm's ROLE inside it; <code>DEGENERATE</code> at <code>famrole_n_members==1</code></div>                                              |
| <b>famrole_class</b> <div><div>ARM</div><div>CATEGORICAL</div><div>100%</div></div>                  | The arm's role in its seed family <b>sole on 1,639 of 2,450 (66.9%)</b> , dominant 282, co and the rest below.                                                                                                                                       |
| <b>famrole_is_dominant</b> <div><div>ARM</div><div>BOOLEAN</div><div>91%</div></div>                 | <i>described by the block above</i> <code>1,779</code> <code>true</code> / <code>452</code> <code>false</code> <div>Defined on: arms with a seed family the arm's ROLE inside it; <code>DEGENERATE</code> at <code>famrole_n_members==1</code></div> |
| <b>famrole_n_members</b> <div><div>ARM</div><div>COUNT</div><div>100%</div></div>                    | <i>described by the block above</i> <code>1</code> <code>22</code> <div>Defined on: arms with a seed family the arm's ROLE inside it; <code>DEGENERATE</code> at <code>famrole_n_members==1</code></div>                                             |
| <b>famrole_var_rank</b> <div><div>ARM</div><div>COUNT</div><div>88%</div></div>                      | <i>described by the block above</i> <code>1</code> <code>14</code> <div>Defined on: arms with a seed family the arm's ROLE inside it; <code>DEGENERATE</code> at <code>famrole_n_members==1</code></div>                                             |

**SURV\_ 8**  
reference context, provenance and rarely-quoted detail

TCGA outcome panel legs.

**8** columns `p-value` **6%** (median 6%) **6** share the block description  
4× `p-value` 3× `real` `0` `1` constant

|                                                                                             |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>surv_dfi_hr_adj</b> <div><div>ARM</div><div>REAL <code>0</code></div><div>6%</div></div> | Adjusted DFI hazard ratio per arm range <b>[0.84, 1.16]</b> , median <b>1.00</b> . The whole distribution sits within $\pm 16\%$ of no effect. See <code>surv_dfi_q_adj</code> . <div>Defined on: arms in the TCGA outcome panel (145) <code>TCGA</code> legs only; the GSE leg is bare-stem matched</div> |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                               |                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>surv_dfi_hr_base</div> <div><div>ARM</div><div>REAL <math>\hat{A} \neq 0</math></div><div>6%</div></div> | <div>described by the block above <math>\hat{A} \cdot 0.855 \hat{A} \neq 1.16</math></div> <div>Defined on: arms in the TCGA outcome panel (145) <math>\hat{A} \neq</math> TCGA legs only; the GSE leg is bare-stem matched</div>    |
| <div>surv_dfi_p_adj</div> <div><div>ARM</div><div>P-VALUE</div><div>6%</div></div>                            | <div>described by the block above <math>\hat{A} \cdot 0.0741 \hat{A} \neq 0.997</math></div> <div>Defined on: arms in the TCGA outcome panel (145) <math>\hat{A} \neq</math> TCGA legs only; the GSE leg is bare-stem matched</div>  |
| <div>surv_dfi_p_base</div> <div><div>ARM</div><div>P-VALUE</div><div>6%</div></div>                           | <div>described by the block above <math>\hat{A} \cdot 0.0973 \hat{A} \neq 0.998</math></div> <div>Defined on: arms in the TCGA outcome panel (145) <math>\hat{A} \neq</math> TCGA legs only; the GSE leg is bare-stem matched</div>  |
| <div>surv_dfi_q_adj</div> <div><div>ARM</div><div>CONSTANT</div><div>6%</div></div>                           | <div>BH q for the arm's DFI hazard ratio.</div>                                                                                                                                                                                      |
| <div>surv_os_hr_adj</div> <div><div>ARM</div><div>REAL <math>\hat{A} \neq 0</math></div><div>6%</div></div>   | <div>described by the block above <math>\hat{A} \cdot 0.816 \hat{A} \neq 1.24</math></div> <div>Defined on: arms in the TCGA outcome panel (145) <math>\hat{A} \neq</math> TCGA legs only; the GSE leg is bare-stem matched</div>    |
| <div>surv_os_p_adj</div> <div><div>ARM</div><div>P-VALUE</div><div>6%</div></div>                             | <div>described by the block above <math>\hat{A} \cdot 0.00235 \hat{A} \neq 0.998</math></div> <div>Defined on: arms in the TCGA outcome panel (145) <math>\hat{A} \neq</math> TCGA legs only; the GSE leg is bare-stem matched</div> |
| <div>surv_os_q_adj</div> <div><div>ARM</div><div>P-VALUE</div><div>6%</div></div>                             | <div>described by the block above <math>\hat{A} \cdot 0.341 \hat{A} \neq 0.998</math></div> <div>Defined on: arms in the TCGA outcome panel (145) <math>\hat{A} \neq</math> TCGA legs only; the GSE leg is bare-stem matched</div>   |

AID\_  $\hat{A} \cdot 4$

reference  $\hat{A} \neq$  context, provenance and rarely-quoted detail

Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates ( `arm_resolvable` , `arm_sep_z` , `oof_drho` medians).

4 columns  $\hat{A} \cdot$  coverage **6% $\hat{A} \neq$ 6%** (median 6%)  $\hat{A} \cdot$  3 share the block description

1x fraction  $\hat{A} \neq 1 \hat{A} \cdot$  1x real  $\hat{A} \neq 0 \hat{A} \cdot$  1x signed  $\hat{A}^{\wedge} 1 \hat{A} \neq 1 \hat{A} \cdot$  1x count

|                                                                                                                                                                            |                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>aid_frac_resolvable</div> <div><div>ARM</div><div>FRACTION <math>\hat{A} \neq 1</math></div></div> <div><div>AGG ARM-IN-FAMILY</div><div>6%</div></div>               | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \neq 1</math></div> <div>Defined on: arms inside <math>\hat{A} \neq 1</math> MULTI-ARM (gene,family) cell <math>\hat{A} \neq</math> the arm rung's 20.3%-of-edges footprint (142)</div>            |
| <div>aid_med_arm_sep_z</div> <div><div>ARM</div><div>REAL <math>\hat{A} \neq 0</math></div></div> <div><div>AGG ARM-IN-FAMILY</div><div>6%</div></div>                     | <div>described by the block above <math>\hat{A} \cdot 0.00108 \hat{A} \neq 5.59</math></div> <div>Defined on: arms inside <math>\hat{A} \neq 1</math> MULTI-ARM (gene,family) cell <math>\hat{A} \neq</math> the arm rung's 20.3%-of-edges footprint (142)</div>   |
| <div>aid_med_oof_drho</div> <div><div>ARM</div><div>SIGNED <math>\hat{A}^{\wedge} 1 \hat{A} \neq 1</math></div></div> <div><div>AGG ARM-IN-FAMILY</div><div>6%</div></div> | <div>described by the block above <math>\hat{A} \cdot -0.092 \hat{A} \neq +0.0421</math></div> <div>Defined on: arms inside <math>\hat{A} \neq 1</math> MULTI-ARM (gene,family) cell <math>\hat{A} \neq</math> the arm rung's 20.3%-of-edges footprint (142)</div> |
| <div>aid_n_cells</div> <div><div>ARM</div><div>COUNT</div></div> <div><div>AGG ARM-IN-FAMILY</div><div>6%</div></div>                                                      | <div>Multi-arm cells this arm participates in <math>\hat{A} \neq</math> fill <b>5.8%</b> (the arm identifiability block is the sparsest on the card).</div>                                                                                                        |

BC\_ 11

reference " context, provenance and rarely-quoted detail

|                                                                                                                                                                                                                                                                         |                                                             |                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Breast-cancer locus annotation for the arm " CpG probes, CGI overlap and related epigenetic context at the precursor locus.<br>11 columns coverage 14%36% (median 14%) 10 share the block description<br>4x fraction 01 4x count 1x boolean 1x signed 11 1x categorical |                                                             |                                                                                                |
| bc_fam_context_homogeneous                                                                                                                                                                                                                                              | described by the block above 720 true / 160 false           | Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)                    |
| bc_fam_dose_coding_host_fraction                                                                                                                                                                                                                                        | described by the block above 0 1                            | Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)                    |
| bc_fam_dose_cotranscribed_fraction                                                                                                                                                                                                                                      | described by the block above 0 1                            | Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)                    |
| bc_fam_n_distinct_coding_hosts                                                                                                                                                                                                                                          | described by the block above 0 3                            | Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)                    |
| bc_fam_n_members                                                                                                                                                                                                                                                        | described by the block above 1 10                           | Defined on: BROADCAST from the seed FAMILY (genomic_context.family_context)                    |
| bc_meth_dbeta                                                                                                                                                                                                                                                           | described by the block above -0.318 +0.226                  | Defined on: BROADCAST from the pri-miRNA HAIRPIN 5p and 3p share one promoter and one 354 arms |
| bc_meth_direction                                                                                                                                                                                                                                                       | Tumour-vs-normal methylation direction at this arm's locus. |                                                                                                |
| bc_meth_n_cgi                                                                                                                                                                                                                                                           | described by the block above 0 38                           | Defined on: BROADCAST from the pri-miRNA HAIRPIN 5p and 3p share one promoter and one 354 arms |
| bc_meth_n_probes                                                                                                                                                                                                                                                        | described by the block above 1 38                           | Defined on: BROADCAST from the pri-miRNA HAIRPIN 5p and 3p share one promoter and one 354 arms |
| bc_meth_normal_beta                                                                                                                                                                                                                                                     | described by the block above 0.0428 0.962                   | Defined on: BROADCAST from the pri-miRNA HAIRPIN 5p and 3p share one promoter and one 354 arms |
| bc_meth_tumour_beta                                                                                                                                                                                                                                                     | described by the block above 0.042 0.954                    | Defined on: BROADCAST from the pri-miRNA HAIRPIN 5p and 3p share one promoter and one 354 arms |

WSHIFT\_ 6

reference " context, provenance and rarely-quoted detail

|                                                                                                                                                                                             |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Within-patient paired quantities on the ~103 matched tumour/NAT participants.<br>6 columns coverage 45%91% (median 52%) 6 share the block description<br>3x real 0 2x real, signed 1x count |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

|                                                                                                                                 |                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>wshift_mean_dGlobalRank</b> <div><div>ARM</div><div>REAL, SIGNED</div><div>52%</div></div>                                   | <i>described by the block above</i> $\hat{A} \cdot -36.2 \hat{=} 42.7$ <div>Defined on: arms in the WITHIN-PATIENT paired subset (<math>n \hat{=} 103</math> matched tumour/NAT pairs; 2,236 arms)</div> |
| <b>wshift_mean_own_shift</b> <div><div>ARM</div><div>REAL, SIGNED</div><div>52%</div></div>                                     | <i>described by the block above</i> $\hat{A} \cdot -3.94 \hat{=} 5.05$ <div>Defined on: arms in the WITHIN-PATIENT paired subset (<math>n \hat{=} 103</math> matched tumour/NAT pairs; 2,236 arms)</div> |
| <b>wshift_n_pairs</b> <div><div>ARM</div><div>COUNT</div><div>91%</div></div>                                                   | <i>described by the block above</i> $\hat{A} \cdot 0 \hat{=} 103$ <div>Defined on: arms in the WITHIN-PATIENT paired subset (<math>n \hat{=} 103</math> matched tumour/NAT pairs; 2,236 arms)</div>      |
| <b>wshift_nat_patient_sd</b> <div><div>ARM</div><div>REAL <math>\hat{A} \cdot \frac{1}{2} 0</math></div><div>74%</div></div>    | <i>described by the block above</i> $\hat{A} \cdot 0 \hat{=} 2.73$ <div>Defined on: arms in the WITHIN-PATIENT paired subset (<math>n \hat{=} 103</math> matched tumour/NAT pairs; 2,236 arms)</div>     |
| <b>wshift_own_specific_frac</b> <div><div>ARM</div><div>REAL <math>\hat{A} \cdot \frac{1}{2} 0</math></div><div>45%</div></div> | <i>described by the block above</i> $\hat{A} \cdot 0 \hat{=} 2.99e+03$ <div>Defined on: arms in the WITHIN-PATIENT paired subset (<math>n \hat{=} 103</math> matched tumour/NAT pairs; 2,236 arms)</div> |
| <b>wshift_sd_dGlobalRank</b> <div><div>ARM</div><div>REAL <math>\hat{A} \cdot \frac{1}{2} 0</math></div><div>45%</div></div>    | <i>described by the block above</i> $\hat{A} \cdot 0.03 \hat{=} 50.9$ <div>Defined on: arms in the WITHIN-PATIENT paired subset (<math>n \hat{=} 103</math> matched tumour/NAT pairs; 2,236 arms)</div>  |

**TIER\_  $\hat{A} \cdot 5$** 

reference  $\hat{A} \cdot$  context, provenance and rarely-quoted detail

Expression tier of the arm against the detection floor  $\hat{A} \cdot$  class, fraction of samples expressed, p90 and max log2 RPM.

5

 columns  $\hat{A} \cdot$  coverage **91% $\hat{A} \cdot$ 91%** (median 91%)  $\hat{A} \cdot$  **4** share the block description

2 $\times$  real  $\hat{A} \cdot \frac{1}{2} 0$   $\hat{A} \cdot$  1 $\times$  categorical  $\hat{A} \cdot$  1 $\times$  boolean  $\hat{A} \cdot$  1 $\times$  fraction 0 $\hat{A} \cdot$ 1

|                                                                                                                    |                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>tier_class</b> <div><div>ARM</div><div>CATEGORICAL</div><div>91%</div></div>                                    | Expression tier $\hat{A} \cdot$ <b>silent on 1,518 of 2,236 arms (67.9%)</b> , conditional 476, the rest robust.                                          |
| <b>tier_expressed</b> <div><div>ARM</div><div>BOOLEAN</div><div>91%</div></div>                                    | <i>described by the block above</i> $\hat{A} \cdot 718$ true / 1,518 false <div>Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)</div> |
| <b>tier_frac_expressed</b> <div><div>ARM</div><div>FRACTION 0<math>\hat{A} \cdot</math>1</div><div>91%</div></div> | <i>described by the block above</i> $\hat{A} \cdot 0 \hat{=} 1$ <div>Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)</div>            |
| <b>tier_max</b> <div><div>ARM</div><div>REAL <math>\hat{A} \cdot \frac{1}{2} 0</math></div><div>91%</div></div>    | <i>described by the block above</i> $\hat{A} \cdot 0.105 \hat{=} 19.4$ <div>Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)</div>     |
| <b>tier_p90</b> <div><div>ARM</div><div>REAL <math>\hat{A} \cdot \frac{1}{2} 0</math></div><div>91%</div></div>    | <i>described by the block above</i> $\hat{A} \cdot 0.105 \hat{=} 18.5$ <div>Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)</div>     |

**DISC\_  $\hat{A} \cdot 2$** 

reference  $\hat{A} \cdot$  context, provenance and rarely-quoted detail

2 columns  $\hat{A} \cdot$  coverage **1% $\hat{A} \cdot$ 100%** (median 50%)

1 $\times$  categorical  $\hat{A} \cdot$  1 $\times$  count

|                                                                                        |                                                                                                                                         |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>disc_gold_families</b> <div><div>ARM</div><div>CATEGORICAL</div><div>1%</div></div> | Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

disc\_n\_gold\_edges

ARM

COUNT

100%

How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p 39 Å· miR-20a-5p 25 Å· miR-93-5p 25 Å· miR-19a-3p 19 Å· miR-18a-5p 9.

CNV\_ Å· 9

reference Å·” context, provenance and rarely-quoted detail

Copy number at the arm's own locus Å·” mean CN and amplification/deletion summaries.  
9 columns Å· coverage 35%Å·35% (median 35%) Å· 8 share the block description  
6× percent Å· 3× real Å·%Å· 0

|                                                                     |                                                                                                                                                                                   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cnv_mean_cn <div>ARM</div> <div>REAL Å·%Å· 0</div> <div>35%</div>   | <div>described by the block above Å· 2.07 Å·” 4.87</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>  |
| cnv_median_cn <div>ARM</div> <div>REAL Å·%Å· 0</div> <div>35%</div> | <div>described by the block above Å· 2 Å·” 4</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>        |
| cnv_pct_altered <div>ARM</div> <div>PERCENT</div> <div>35%</div>    | <div>described by the block above Å· 29.6 Å·” 81.8</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>  |
| cnv_pct_amp <div>ARM</div> <div>PERCENT</div> <div>35%</div>        | <div>described by the block above Å· 8.58 Å·” 63</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>    |
| cnv_pct_deep_del <div>ARM</div> <div>PERCENT</div> <div>35%</div>   | <div>Percent of samples with a deep deletion at this arm's locus.</div>                                                                                                           |
| cnv_pct_gain <div>ARM</div> <div>PERCENT</div> <div>35%</div>       | <div>described by the block above Å· 9.34 Å·” 23</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>    |
| cnv_pct_loh <div>ARM</div> <div>PERCENT</div> <div>35%</div>        | <div>described by the block above Å· 2.67 Å·” 72.3</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>  |
| cnv_pct_loss <div>ARM</div> <div>PERCENT</div> <div>35%</div>       | <div>described by the block above Å· 0 Å·” 31.2</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div>     |
| cnv_sd_cn <div>ARM</div> <div>REAL Å·%Å· 0</div> <div>35%</div>     | <div>described by the block above Å· 0.913 Å·” 4.13</div> <div>Defined on: arms with a mapped CN locus in the cohort stratum (856) Å·” Å· multi-locus arms are DILUTED here</div> |

GCTX\_ Å· 16

reference Å·” context, provenance and rarely-quoted detail

Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware.  
16 columns Å· coverage 24%Å·29% (median 29%) Å· 15 share the block description  
6× categorical Å· 4× boolean Å· 3× identifier Å· 1× constant Å· 1× count Å· 1× real Å·%Å· 0

|                                                                 |                                                                                                                                                                                       |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gctx_clustered <div>ARM</div> <div>BOOLEAN</div> <div>29%</div> | <div>described by the block above Å· 310 true / 404 false</div> <div>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> Å·” 714)</div> |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                              |                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>gctx_context_mixed</div><div><div>ARM</div><div>BOOLEAN</div><div>29%</div></div></div>            | <div>described by the block above Â·20 true / 694 false</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>  |
| <div><div>gctx_coord_source</div><div><div>ARM</div><div>CONSTANT</div><div>29%</div></div></div>            | <div>Provenance of the genomic-context coordinates.</div> <div></div>                                                                                                     |
| <div><div>gctx_host</div><div><div>ARM</div><div>IDENTIFIER</div><div>27%</div></div></div>                  | <div>described by the block above Â·356 distinct</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>         |
| <div><div>gctx_host_region</div><div><div>ARM</div><div>IDENTIFIER</div><div>27%</div></div></div>           | <div>described by the block above Â·74 distinct</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>          |
| <div><div>gctx_host_type</div><div><div>ARM</div><div>CATEGORICAL</div><div>27%</div></div></div>            | <div>described by the block above Â·7 levels</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>             |
| <div><div>gctx_in_exon</div><div><div>ARM</div><div>BOOLEAN</div><div>29%</div></div></div>                  | <div>described by the block above Â·218 true / 496 false</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div> |
| <div><div>gctx_mir_class</div><div><div>ARM</div><div>CATEGORICAL</div><div>29%</div></div></div>            | <div>described by the block above Â·4 levels</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>             |
| <div><div>gctx_mirgenedb</div><div><div>ARM</div><div>BOOLEAN</div><div>29%</div></div></div>                | <div>described by the block above Â·526 true / 188 false</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div> |
| <div><div>gctx_n_loci</div><div><div>ARM</div><div>COUNT</div><div>29%</div></div></div>                     | <div>described by the block above Â·1 â€” 3</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>              |
| <div><div>gctx_promoter_class</div><div><div>ARM</div><div>CATEGORICAL</div><div>29%</div></div></div>       | <div>described by the block above Â·4 levels</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>             |
| <div><div>gctx_promoter_coord_class</div><div><div>ARM</div><div>CATEGORICAL</div><div>29%</div></div></div> | <div>described by the block above Â·4 levels</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>             |
| <div><div>gctx_promoter_gene</div><div><div>ARM</div><div>IDENTIFIER</div><div>24%</div></div></div>         | <div>described by the block above Â·307 distinct</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>         |
| <div><div>gctx_promoter_gene_type</div><div><div>ARM</div><div>CATEGORICAL</div><div>24%</div></div></div>   | <div>described by the block above Â·5 levels</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>             |
| <div><div>gctx_promoter_host_tss_kb</div><div><div>ARM</div><div>REAL Â·0</div><div>24%</div></div></div>    | <div>described by the block above Â·0 â€” 874</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>            |
| <div><div>gctx_strand_rel</div><div><div>ARM</div><div>CATEGORICAL</div><div>27%</div></div></div>           | <div>described by the block above Â·2 levels</div> <div>  Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€™ 714)</div>             |

**SURROGATE\_ Â· 2**

reference â€™ context, provenance and rarely-quoted detail

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for.

2 columns Â· coverage 0â€™0% (median 0%) Â· 2 share the block description

1x fraction 0â€™1 Â· 1x categorical

|                                                                                                     |                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>surrogate_corr</div><div><div>ARM</div><div>FRACTION 0â€™1</div><div>0%</div></div></div> | <div>described by the block above Â·0.621 â€” 0.892</div> <div>  Defined on: arms with a same-seed SURROGATE (MH-166) â€™ 3.6% of edges</div> |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

surrogate\_instrument

described by the block above 4 levels

ARM

CATEGORICAL

0%

Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges

CNVC\_ 5

reference context, provenance and rarely-quoted detail

CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.

5 columns coverage 31%31% (median 31%) 4 share the block description

2x p-value 2x signed 11 1x count

cnvc\_n

described by the block above 20 1032

ARM

COUNT

31%

Defined on: arms with a CN expression concordance fit (756)

cnvc\_partial\_p

described by the block above 1.25e-42 1

ARM

P-VALUE

31%

Defined on: arms with a CN expression concordance fit (756)

cnvc\_partial\_q

BH q for the arm's CN-coupling partial correlation.

ARM

P-VALUE

31%

cnvc\_partial\_rho

described by the block above -0.322 0.468

ARM

SIGNED 11

31%

Defined on: arms with a CN expression concordance fit (756)

cnvc\_spearman\_rho

described by the block above -0.378 0.5

ARM

SIGNED 11

31%

Defined on: arms with a CN expression concordance fit (756)

FOC\_ 7

reference context, provenance and rarely-quoted detail

Paralog-decontaminated FOCAL-locus concordance.

7 columns coverage 30%30% (median 30%) 7 share the block description

2x count 2x signed 11 1x identifier 1x boolean 1x p-value

foc\_focal\_locus

described by the block above 437 distinct

ARM

IDENTIFIER

30%

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) 8 DESCRIPTIVE, not an instrument

foc\_multilocus

described by the block above 35 true / 698 false

ARM

BOOLEAN

30%

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) 8 DESCRIPTIVE, not an instrument

foc\_n

described by the block above 41 1032

ARM

COUNT

30%

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) 8 DESCRIPTIVE, not an instrument

foc\_n\_loci

described by the block above 1 3

ARM

COUNT

30%

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) 8 DESCRIPTIVE, not an instrument

foc\_partial\_p

described by the block above 1.54e-42 0.996

ARM

P-VALUE

30%

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) 8 DESCRIPTIVE, not an instrument

foc\_partial\_rho

described by the block above -0.269 0.424

ARM

SIGNED 11

30%

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) 8 DESCRIPTIVE, not an instrument

foc\_spearman\_rho

ARM

SIGNED -1

30%

described by the block above -0.235 +0.485

Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) DESCRIPTIVE, not an instrument

MODEL\_ 1

reference context, provenance and rarely-quoted detail

1 columns coverage 30% (median 30%)

1x count

model\_n\_genes

Which model variant produced this row.

ARM

COUNT

AGG EDGE

30%

Defined on: arms carrying >=1 curated HE edge in the model design (741)

COMV\_ 2

reference context, provenance and rarely-quoted detail

Co-expression module membership and top co-expressed partner. Pure co-expression the cross-state class columns are deliberately excluded from this block.

2 columns coverage 15% (median 15%) 2 share the block description

1x count 1x identifier

comv\_module

described by the block above 1 25

ARM

COUNT

15%

Defined on: arms in a co-expression module (371) pure co-expression; the state-class columns are excluded

comv\_top\_partner

described by the block above 371 distinct

ARM

IDENTIFIER

15%

Defined on: arms in a co-expression module (371) pure co-expression; the state-class columns are excluded

CLOAD\_ 3

reference context, provenance and rarely-quoted detail

Compartment LOADING of the arm.

3 columns coverage 8% (median 60%) 3 share the block description

1x real 0 1x fraction 1 1x categorical

cload\_dose\_rpm

described by the block above 0.279 2.4e+05

ARM

REAL 0

60%

Defined on: arms with a compartment-loading row (1,473) its source has NO PRODUCER (MH-111 ad-hoc)

cload\_lock

described by the block above 0.0761 0.547

ARM

FRACTION 1

8%

Defined on: arms with a compartment-loading row (1,473) its source has NO PRODUCER (MH-111 ad-hoc)

cload\_top\_compartment

described by the block above 8 levels

ARM

CATEGORICAL

60%

Defined on: arms with a compartment-loading row (1,473) its source has NO PRODUCER (MH-111 ad-hoc)

HX\_

1

reference " context, provenance and rarely-quoted detail

|    |         |  |          |                     |
|----|---------|--|----------|---------------------|
| 1  | columns |  | coverage | 28%28% (median 28%) |
| 1x | real    |  |          | 0                   |

hx\_hly\_baseline\_imputed

ARM

REAL 0

28%

" IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) " LABELLED as imputed, never presented as a measurement.

seed\_family\_card.tsv

# seed\_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS for those every 'family-level' statistic is trivially that one arm's value. Mask on fam\_single\_member first.

33 columns 1,959 rows 2 blocks median coverage 92%

## Contents 2 blocks, most important first

1. seed\_1 CORE
2. fam\_32 REFERENCE

### SEED\_1

core the columns findings rest on

1 columns coverage 100% (median 100%)  
1x identifier

seed\_family KEY the seed family itself, one row per family.

KEY IDENTIFIER 100%

### FAM\_32

reference context, provenance and rarely-quoted detail

Seed-family structure the arm sits in membership, affinity and dominance within it.

32 columns coverage 4% (median 91%) 17 share the block description  
12x count 8x real 7x fraction 1 2x boolean 1x categorical 1x identifier 1x signed  
1

#### fam\_ctx\_composition

FAMILY CATEGORICAL 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.

#### fam\_ctx\_context\_homogeneous

FAMILY BOOLEAN 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.

#### fam\_ctx\_dose\_coding\_host\_frac

FAMILY FRACTION 1 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.

#### fam\_ctx\_dose\_cotranscribed\_frac

FAMILY FRACTION 1 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.

#### fam\_ctx\_n\_distinct\_coding\_hosts

FAMILY COUNT 29%

The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.

|                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>fam_cur_degree_max</div><div><div>FAMILY</div><div>COUNT</div><div>36%</div></div></div>                                           | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 112</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                            |
| <div><div>fam_dominant_arm</div><div><div>FAMILY</div><div>IDENTIFIER</div><div>92%</div></div></div>                                        | <div>described by the block above <math>\hat{A} \cdot 1,810</math> distinct</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                                      |
| <div><div>fam_dominant_share</div><div><div>FAMILY</div><div>FRACTION <math>0 \hat{A} \epsilon'' 1</math></div><div>92%</div></div></div>    | The top member's share of the family's total dose.                                                                                                                                                                                                                                                                                                                                      |
| <div><div>fam_dose_hhi</div><div><div>FAMILY</div><div>FRACTION <math>0 \hat{A} \epsilon'' 1</math></div><div>15%</div></div></div>          | Concentration of DOSE across the family's members. $\hat{a} \epsilon \dots$ <b>FLOOR-CORRECTED <math>\hat{a} \epsilon''</math> verified 2026-08-19</b> : its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = <b>+0.1896</b> rather than the strongly negative value a raw index would give. |
| <div><div>fam_dose_max_log2</div><div><div>FAMILY</div><div>REAL <math>\hat{A} \epsilon \text{¥} 0</math></div><div>92%</div></div></div>    | <div>described by the block above <math>\hat{A} \cdot 0.105 \hat{A} \epsilon''</math> 17.9</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                       |
| <div><div>fam_dose_med_log2</div><div><div>FAMILY</div><div>REAL <math>\hat{A} \epsilon \text{¥} 0</math></div><div>92%</div></div></div>    | <div>described by the block above <math>\hat{A} \cdot 0.105 \hat{A} \epsilon''</math> 16</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                         |
| <div><div>fam_dose_total_log2</div><div><div>FAMILY</div><div>REAL <math>\hat{A} \epsilon \text{¥} 0</math></div><div>100%</div></div></div> | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 17.9</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                           |
| <div><div>fam_fame_npmid_sum</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                          | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 577</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                            |
| <div><div>fam_n_expressed</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 14</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                             |
| <div><div>fam_n_guide</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                                 | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 9</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                              |
| <div><div>fam_n_members</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                               | Arms in the seed family (median 1, max 22).                                                                                                                                                                                                                                                                                                                                             |
| <div><div>fam_n_members_context</div><div><div>FAMILY</div><div>COUNT</div><div>29%</div></div></div>                                        | Arms in the seed family (median 1, max 22).                                                                                                                                                                                                                                                                                                                                             |
| <div><div>fam_n_model_edges</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                           | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 209</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                            |
| <div><div>fam_n_passenger</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                             | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 3</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                              |
| <div><div>fam_n_shift_gt20_gated</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>                                      | <div>described by the block above <math>\hat{A} \cdot 0 \hat{A} \epsilon''</math> 2</div> <div>Defined on: seed families in the arm-card universe (1,959; <math>\hat{a}</math>' 83.7% single-member <math>\hat{a}^+</math>' shares/HHI are 1 or NaN)</div>                                                                                                                              |

|                                                                                                                |                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>fam_real_n_c10_sum</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>            | <div><div>described by the block above</div><div>0.115</div><div>Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)</div></div>                                                  |
| <div><div>fam_real_n_scored_sum</div><div><div>FAMILY</div><div>COUNT</div><div>100%</div></div></div>         | <div><div>described by the block above</div><div>0.169</div><div>Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)</div></div>                                                  |
| <div><div>fam_redund_m_eff</div><div><div>FAMILY</div><div>REAL 0</div><div>4%</div></div></div>               | <div>Within-family redundancy effective member count, over-count fraction, and the median within-family correlation.</div>                                                                                                            |
| <div><div>fam_redund_median_within_rho</div><div><div>FAMILY</div><div>SIGNED 1</div><div>4%</div></div></div> | <div>Within-family redundancy effective member count, over-count fraction, and the median within-family correlation.</div>                                                                                                            |
| <div><div>fam_redund_overcount_frac</div><div><div>FAMILY</div><div>FRACTION 1</div><div>4%</div></div></div>  | <div>Within-family redundancy effective member count, over-count fraction, and the median within-family correlation.</div>                                                                                                            |
| <div><div>fam_seed_shift_max_gated</div><div><div>FAMILY</div><div>FRACTION 1</div><div>13%</div></div></div>  | <div><div>described by the block above</div><div>0.0010.645</div><div>Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)</div></div>                                             |
| <div><div>fam_single_member</div><div><div>FAMILY</div><div>BOOLEAN</div><div>100%</div></div></div>           | <div>THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS. For those rows every 'family-level' statistic is TRIVIALY that one arm's value, and fam_dominant_share is forced to exactly 1.0.</div> |
| <div><div>fam_targetome_seed_med</div><div><div>FAMILY</div><div>REAL 0</div><div>97%</div></div></div>        | <div>Family-level targetome size.</div>                                                                                                                                                                                               |
| <div><div>fam_targetome_seq_med</div><div><div>FAMILY</div><div>REAL 0</div><div>30%</div></div></div>         | <div>Family-level targetome size.</div>                                                                                                                                                                                               |
| <div><div>fam_var_hhi</div><div><div>FAMILY</div><div>FRACTION 1</div><div>14%</div></div></div>               | <div><div>described by the block above</div><div>2.41e-080.926</div><div>Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)</div></div>                                          |
| <div><div>fam_var_max</div><div><div>FAMILY</div><div>REAL 0</div><div>89%</div></div></div>                   | <div><div>described by the block above</div><div>0.005453.11</div><div>Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)</div></div>                                            |
| <div><div>fam_var_med</div><div><div>FAMILY</div><div>REAL 0</div><div>89%</div></div></div>                   | <div><div>described by the block above</div><div>0.005452.84</div><div>Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)</div></div>                                            |